Heredity: Module 5

Sexual and Asexual Reproduction:

Sexual Reproduction:

- 2 parents
- Genetic material from both parents combine
  - Produce offspring
- Every species has set number of chromosomes in each somatic cell
  - Humans:
    - 46
    - 23 from mom, 23 from dad
- Chromosomes occur in homologous pairs
  - Homologous pairs:
    - Same genes in same order
      - Different alleles (variants of gene)
- Chromosome number not change across generations
- Meiosis
  - Cell division process
  - Ensures each parent contributes 1/2 of their chromosomes to offspring
    - Prevents doubling across generations
      - As chromosomes are put in by both parents
  - Happens in reproductive organs
    - Plants, animals
  - Produces gametes (sex cells)
    - Has one set of chromosomes
      - 23 chromosomes instead of 46
- Diploid:
  - 2n
  - 2 sets of chromosomes
  - Human somatic (body) cells
    - 46 chromosomes
- Haploid:
  - n
  - 1 set of chromosomes
  - Gametes

        - 23 chromosomes
- n = 1 set of chromosomes
    - 23 chromosomes in gametes
- 2n = 2 sets of chromosomes
    - 46 chromosomes in somatic cells
- Offspring inherit 1 set of chromosomes from
    - Mom
        - Maternal chromosomes
    - Dad
        - Paternal chromosomes
- n (egg) + n (sperm) = 2n (zygote)
- Fertilised egg
    - Zygote (diploid)
        - Divides using mitosis
            - Becomes embryo
            - With genetically identical diploid body cells
- Di in diploid → two

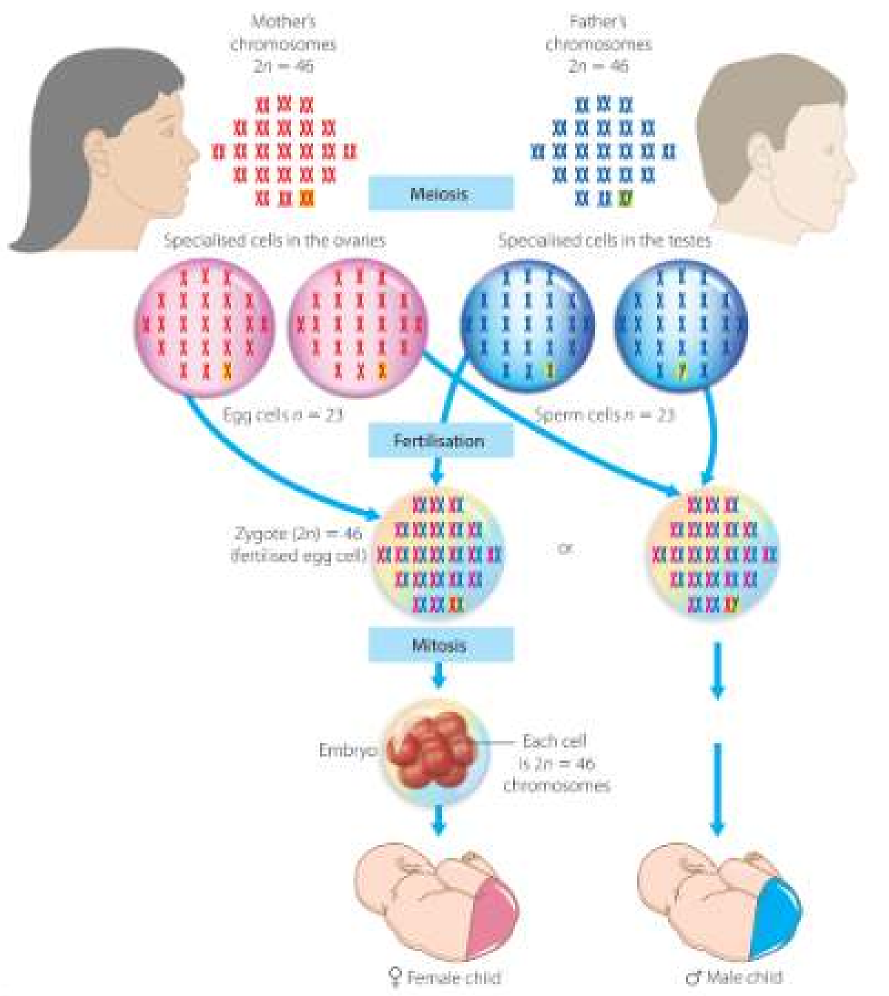

FIGURE 2.3 Maintenance of the diploid number during sexual reproduction in humans

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Advantages and Disadvantages:

Advantages:	Disadvantages:

- Offspring have mix of genes from both parents
    - Are not genetically identical to:
        - parents
        - Other offspring (siblings)
- Genetic variation
- Genetic diversity
    - For continuity of species
- Some offspring have random variations
    - Better suited to new, changing environmental conditions
	- More
	    - Time
	    - Energy
	    - Resources needed
	- Processes like
	    - Finding mate
	    - Courtship behaviour
	    - Gamete production
	    - Mating
	- Processes may make organisms more vulnerable to predators

- May outcompete other individuals in species
  - Selective advantage
- When environmental change
  - Survival of some individuals
    - Give species better chance for survival
      - If advantageous genes passed on	

Animals:

- Most are unisexual
  - One sex
    - Male
    - Female
- Hermaphrodites
  - Are bisexual
    - Have male + female reproductive organs
- Hermaphroditism:
  - Advantages:
    - Good for species with low population
    - Non moving animals
      - Finding mate hard
  - Disadvantages:
    - Individuals spend more energy
      - To grow + maintain
        - 2 sets of reproductive organs
    - If self fertilisation:
      - Gametes, offspring have little genetic variation
- Other reproductive strategies:
  - Type of fertilisation
    - Internal external
  - Number of gametes produced
  - Timing of gametes release
  - Where young develop
    - Outside

    - Inside body
  - Nature of parental care

External and Internal Fertilisation:

- Successful fertilisation:
  - Sperm + ova
    - Must meet
    - Not dehydrate

External:

- fertilisation
  - Outside body
- Better for organisms
  - in aquatic environment
    - Marine animals
  - In moist environment
    - Earthworms
- Success chances increased by
  - Sync of
    - Reproductive cycles
    - Mating behaviours
    - Release of gametes
- little, no parental care
- Less time, energy needed from parents
- Larger number of gametes
  - Need to be produced
- Advantage:
  - Wide dispersal of young
  - Reduces competition for food, living space
  - Rapid recovery of populations away from damaged areas

Internal:

- fertilisation
  - inside female body
- Terrestrial organisms
  - Insects
  - Lizards

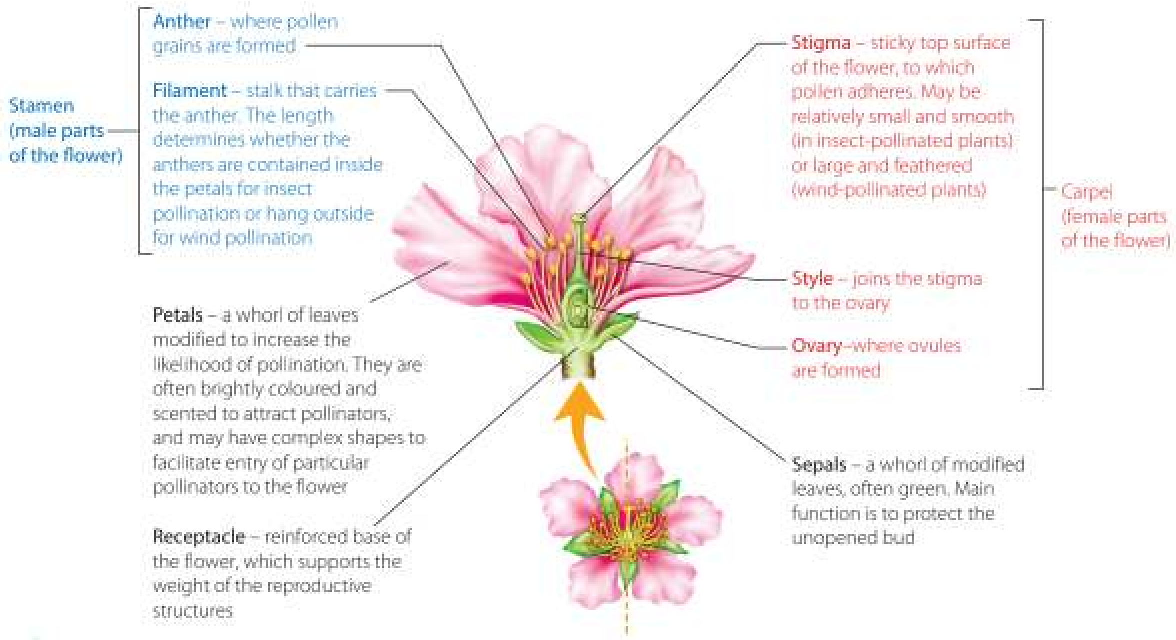
    - Kangaroos
    - Humans
- Vertebrate reproduction
    - Started in
        - Ocean
            - fish
        - Freshwater environments
            - Amphibians
    - Changed when reptiles, birds, mammals
        - Colonised land, air
- Internal environment:
    - Protects gametes + offspring from
        - Dehydration
        - External elements
        - Immediate predation
- Less eggs needed for enough offspring survival
- Zygote → embryo
    - In moms body
    - Gets nutrients from placenta
    - Born alive

TABLE 2.1 Comparison of internal and external fertilisation in animals

CHARACTERISTICS	DIFFERENCES		SIMILARITIES
	EXTERNAL FERTILISATION	INTERNAL FERTILISATION	
Gametes	Large numbers of male and female gametes produced	Large number of male gametes and fewer female gametes produced	Male and female gametes required – sperm and eggs (ova)
Union	Occurs in open water environments	Occurs inside the reproductive tract of the female in organisms that live mostly or completely on land	Sperm fertilise the eggs when they unite
Conception mechanism	Simultaneous release of gametes	Copulation: the male inserts sperm into the female's reproductive tract via penis or cloaca	Sperm will fertilise eggs when in very close proximity to each other; gametes require a watery environment for this to occur
Chance of fertilisation	Low, because male gametes are released into a large open area where there is less chance of successfully uniting with female gametes	High, because male gametes are released into a confined space where there is more chance of successfully uniting with female gametes	If male and female gametes are in close proximity to each other, fertilisation will usually occur
Environment for zygote	Usually external, in a watery environment that is vulnerable to environmental elements such as temperature, predation, infection and rapid dispersal from the area	Usually internal, in a very protected environment inside the female's body. Temperature is controlled and there is less chance of predation, infection and loss of zygote from the area	Zygote requires a watery environment for development
Number of offspring/zygotes	Usually a larger number than in internal fertilisation, but many zygotes perish and so a smaller number of offspring survive	A smaller number of offspring than in external fertilisation, because very few perish (higher success rate)	Zygote number is determined by the number of sperm and ova that successfully fuse
Breeding frequency	More frequent than in internal fertilisation due to the lower fertilisation success rate	Seasonal and less frequent than in external fertilisation due to higher fertilisation success rate and greater energy costs	Breeding frequency depends on the requirements of the species and the favourability of environmental conditions
Parental investment	Usually no parental care	Parental care of eggs and/or developing young is more common	Parental investment is indirectly proportional to the number of gametes produced

Plants:

- Success in fertilisation
- Female parts
    - Carpel
    - Gynoecium
- Male parts
    - Stamen
- Both male, female parts
- Non sexual parts
    - Petals
    - Sepals



- For fertilisation:
    - Male gametes
        - In pollen
        - Carried from anthers
            - To female part of flower
                - Stigma
                    - Pollen tube that
                    - Germinates, grows down the style
        - Pollen tube carries it to ovule
            - In ovary
- Internal fertilisation

Reproductive strategies:

- External agents carry gametes
    - From one parent to another
        - "Pollinating agents"
- Pollinating agents
    - carry plant zygote far from parent
- Increase survival chances in harsh environments
- Evolution of flowers
    - Better pollination
    - More diversity in flowering plants

Process of seed production, pollination, fertilisation:

1. Meiosis:
    - Occurs in
        - Anther
        - Ovary
    - Produce haploid gametes:
        - Anther: sperm
        - Ovary: egg
2. Pollination:
    - Pollen grains transferred to stigma
    - Occur via:
        - Wind
        - Animals
        - Other pollinators
3. Fertilisation:
    - Pollen grain germinates
        - Grows pollen tube down style
    - Male gamete travels
        - Through tube → egg in ovule
    - Fertilisation occurs
4. Embryo + seed formation:
    - Diploid zygote → embryo
    - Ovule
        - → seed
        - Has embryo
5. Seed dispersal:
    - Seed released from parent plant
        - Dispersed by
            - Wind
            - animals
6. Germination:
    - In suitable conditions
    - Seed germinates
    - Embryo grows
7. Mitosis:
    - Embryo grows → mature plant

- Produces more diploid cells

Self Pollination:

- Less energy
- No need for
    - Showy petals
    - Nectar

      To attract other pollinators
- Common in areas with no pollinators

Cross Pollination:

- Needs external agents
    - Wind
    - Water
    - Insects
    - Birds
    - Mammals
- Leads to More genetic variation
- As flowers become specialised
    - Pollinating agents more specialised

Pollination:

By Wind:

- Flowers are
    - Small
    - Greenish
    - Odourless
    - With reduced petals
- Found in
    - Grass
    - Many angiosperms
- Anthers
    - Produce large amounts of light loon
- Stigmas
    - Large and feathery
- Is inefficient
- High pollen production needed

- Pollen grain structure
    - Helps maintain species compatibility

By Animals:

- Flowers that attract animals
    - More effective in
        - ensuring transfer of pollen
- 1-1 relo with plant + animal
    - Reduces waste of pollen
        - Ensuring its put onto correct flower
- Animals:
    - Insects
    - Birds
    - Mammals
    - Search for reward nectar
    - Provided by pollen or flower nectaries
- Pollen rubs onto body of pollinator
    - Going to next flower they go to
- Flower:
    - Scent
    - Colour
    - Markings
    - Shape
    - Nectar

    Important for attracting animals

Seed Dispersal:

- Seeds dispersed
- Advantage
    - For seeds dispersed over wide distance
        - Helps prevent
            - Overcrowding
            - Competition for light water soil nutrients
        - Increases continuity of species
            - In other regions
                - In case of change in local environment
- Relies on

    - Type of fruit in which seeds occur
    - Match type of dispersal agent in environment
- Dispersal agent
    - Animal
    - Water
    - air

Asexual Reproduction:

- One parent
- No gametes
- One parent passes down
    - All genetic material in offspring
- All offspring
    - Genetically identical to parent
    - To each other
- No genetic variation
- Unicellular organisms do this
- Selective pressures for asexual > sexual reprod.
    - Shortage of food, other resources
        - Less energy use
    - Small mating population
    - Time, other constraints on finding mate
    - Protists, fungi, plants
        - Alternate between
            - Sexual
            - Asexual reprod.
            - In their life cycle
        - Alternation of generations
            - 1 gen sexual
            - Next gen asexual

Advantages and Disadvantages:

Advantages:	Disadvantages:

- Quickly reproduce
    - Without finding mate
	- Little → no genetic variation
	    - All species

- Good for immobile organisms
    - Plants
- Good for harsh environments
- Well adapted organisms to their environment
- Selective pressures for asexual > sexual reprod.
    - Shortage of food, other resources
        - Less energy use
    - Small mating population
    - Time, other constraints on finding mate
	- Vulnerable to sudden change in environment
	- Result in survival of only few individuals

Plants:

zVegetative Propagation:

- Vegetative organs:
    - Bulbs
    - Tubers
    - Rhizomes
    - suckers
- Offspring
    - Clones of parent
- Tuber, bulbs
    - Store food
    - Survive through extreme conditions
        - Cold
        - Drought
- Organs regrow when conditions improve
- Used in agriculture
    - To seedless or
    - Hard-to-germinate crops

Types of Vegetative Structures:

- Runners - modified stems:
    - Long thin stems
        - Growing on soil surface

    - In strawberries, spinifex grass
    - Make new plants at nodes
        - By forming roots + shoots
- Rhizomes - modified stems:
    - Horizontal underground stems
    - In ginger, ferns, bracken
    - Split to grow new plants
- Suckers - modified roots:
    - New shoots from roots
    - In plants like
        - Reeds
        - Wattles
    - Good for rapid regrowth after disturbance
- Apoximis:
    - From generative tissues
    - May include
        - Unfertilised ovules, leaf tissue
    - Generative tissue:
        - Makes plantlets that
            - Produce asexual seeds
    - Advantages:
        - Rapid multiplication
        - Plantlets able to produce seeds
            - Increases seed dispersal
    - Includes parthenogenesis

Other Organisms:

Budding:

- Adult organism → small bud
- Bud separates from parent
    - Grows into new individual
        - Genetically identical to parent

Fungi:

- Yeast:
    - Fungi

- Unicellular
- Under favourable conditions:
    - Yeast cells form buds on surface
    - Parent replicate dna
    - Nuclei divides into the buds
    - Buds may
        - Detach, grow separately
        - Grow on parent
            - Form chain of cells
- Multiply rapidly
- Reproduce sexually when
    - Lack of nutrients
- Exist in haploid or diploid form
    - Both reprod. Assexually by budding
- Diploid form:
    - Resistant to harsher conditions
    - Form spores
        - That germinate
        - When conditions favourable

Multicellular Organisms

- Jellyfish
- Hydra

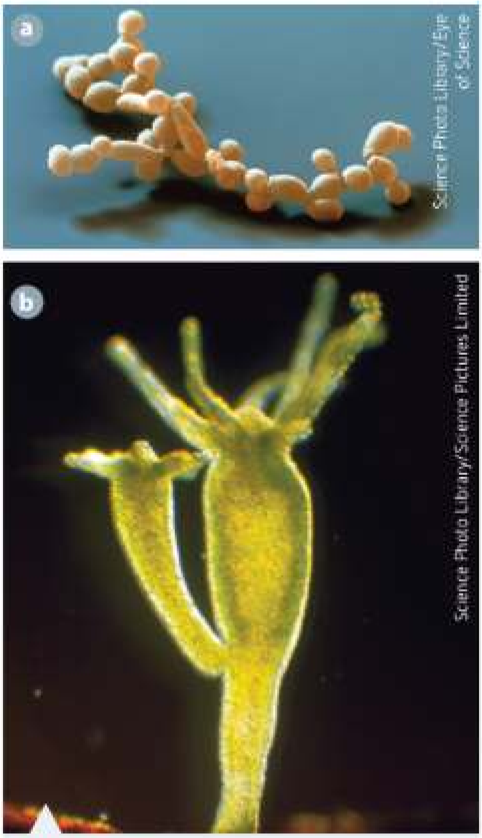
- 

FIGURE 2.25 **a** Budding in yeast, where daughter cells remain attached to the parent cell; **b** hydra budding by producing a multicellular outgrowth from the side of the parent's body

Binary Fission:

- Splitting into 2
- Main method of asexual reprod. In
    - Unicellular organisms
        - Bacteria
            - Prokaryotes
        - Protists
            - eukaryotes
- Cell
    - Grows to x2 its size
    - Replicates genetic material
    - Splits into 2 cells
        - With identical genetic material
- Each individual
    - Must have complete, exact copy of dna
- Advantage:
    - Allows for rapid population growth

    - Over short period of time

Bacteria:

- Cells grows to full adult size
- Replicates single dna molecule
- Each copy of dna
  - Attaches to opposite ends of cell membrane
- Cytoplasm pinched off
- Dna not damaged
- New cell wall made in
  - Area of cell cleavage
- Cell needs to grow full adult size
  - Before division again
- Timing + sequence of steps
  - Controlled in cell cycle
- Research and medicine:
  - Antibiotics that block fission in pathogenic bacteria
- Some bacteria reprod. by
  - Budding
  - Multiple divisions from 1 big cell

Protists:

- Unicellular
- Eukaryotes
- Binary fission Involves mitosis
  - Making of spindle with cell cytoplasm

Spores:

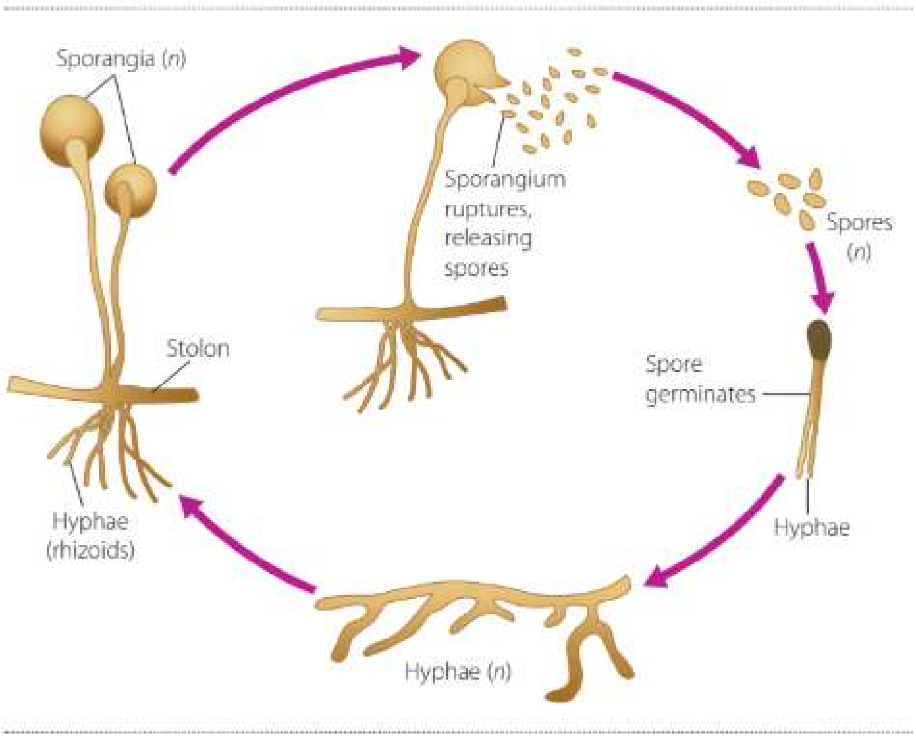
- Small
- Unicellular
- Reproductive cells
- Produced in large numbers
- Organisms:
  - Fungi
  - Some plants
- Sporangia
  - Grow from hyphae

- Make great amounts of spores
  - Spores Light
  - Easily dispersed
  - Travel long distances by wind
- Expand distribution of species
- Not gamete
  - No need to fuse with other cells
- Differ from seeds
  - Each spore is
    - Single cell
    - Not carrying embryo or food

Fungi:

- Multicellular fungi
  - Mould
  - Mushrooms
  - Made up of threads, filaments
    - "Hyphae"
      - Branched
      - Interconnected
      - Forming main body
      - "Mycelium"
- Rhizopus nigricans (black bread mould), mucor, penicillum
- Fungi may have
  - Many shared nuclei
  - In shared cytoplasm
- Under favourable conditions:
  - Form asexual spores via mitosis
- Mature white → black
- Dispersed by air
- Germinating spores
  - Form new mycelium
  - Allow rapid spread

FIGURE 2.30
Asexual reproduction in a fungus such as *Rhizopus* (bread mould)



- Advantages:
    - Rapid reproduction + colonisation of environments
    - Genetically identical to parent
    - Useful in stable environments
    - Needs
        - Less energy
        - Less nutrients
    - Species continuity in familiar habitats
- Sexual reproduction when:
    - Environmental changes
    - 2 mating types of hyphae fuse
        - Forms fruiting body
            - Produce genetically diverse spores
                - Increase survival

Sexual Reproduction in Mammals:

- Mechanisms for reproductive success:
    - Internal fertilisation:
        - All mammals
        - Increase likelihood of gametes meeting
    - Implantation of embryo
        - Marsupials, eutherians

- Into uterine wall
- Internal development of embryo
  - Increases chance of survival
- Pregnancy:
  - Marsupials, eutherians
  - Developing young protected from external environment
  - Have nutrient supply
  - Complete gestation period
- Whole reproduction - gamete → adulthood → offspring
  - Regulated by hormones

Hormones:

- Chemical messengers
  - Coordinate body functions
    - Reproduction
- Pituitary gland
  - Master gland
    - Controls hormone secretion
      - That stimulate or inhibit other endocrine glands
- Sex hormones
  - Regulate
    - Reproductive organ development
    - Secondary sex characteristics
- Puberty activates reproductive system
  - Under hormonal control
- Gonads
  - Ovaries, testicles
  - Start gamete production at puberty
    - Through gametogenesis

Breeding Seasons - Hormones:

- Some mammals
  - Have once or twice a year breeding season
  - Based on female fertility
  - "Seasonal breeders"
  - Sheep, cattle
- Higher order primates:

- All year round breeding season
- Female fertility occurs in cycle
    - Throughout whole year
    - ∴ are sexually active all year round

Mammalian Reproduction - Hormones:

- Androgens:
    - Male hormone
    - Control functioning + development of
        - Male sex organs
        - Secondary sex characteristics
    - Cells in testicles:
        - Release testosterone
            - Plays primary role in
                - Production of sperm (spermatogenesis)
- Oestrogens:
    - Main female hormones
    - Control functioning + development of
        - Female sex organs
        - Secondary sex characteristics
    - Responsible of oestrus (in heat)
        - Just before ovulation
            - In female mammals
            - That are seasonal breeders
    - Main function is
        - Ovarian functioning
        - Fertility in females
- Progestrogens
    - 2nd group of female hormones
    - Progesterone
        - Most common
        - Primary role in pregnancy
        - Lactation
        - Initiates menstruation

Female Reproductive Cycle - Hormones:

- Endocrine glands regulate and coordinate

    - Ovarian
    - Menstrual cycles
    - Coordination increases:
        - Fertility
        - Probability of successful reproduction
        - Biological fitness
        - Continuity of species
- Hormones of pituitary gland + gonads
    - Regulate reproductive cycles in mammals
- Pituitary gland
    - Secretes hormones
        - That Regulate other endocrine glands
            - Ovary of females
    - Secretes 2 gonadotropic hormones (GnRH):
        - Follicle stimulating hormone (FSH)
            - Stimulates maturation of many follicles
                - In ovaries
        - Luteinising hormone (LH)
            - Promotes final maturation of
                - 1 Ovarian follicle
            - Ovulation
            - development of
                - Corpus luteum

Oestrogen and Progesterone:

- made by ovaries
- Controlled by pituitary hormones
- Regulate
    - Ovarian cycle by:
        - controlling production, maturation
            - Of gametes (ova) in ovaries
    - Menstrual cycle by:
        - Preparing uterus for
            - Implantation of fertilised egg
                - Every cycle
        - If no fertilisation
            - Oestrogen + progesterone decrease

            - Lining of uterus tears away
                - → bleeding
                    - Menstruation
    - Maintenance of pregnancy
    - Prep for + maintenance of
        - Lactation

Oestrogen:

- made by:
    - Developing follicles in ovary
- Functions:
    - Stimulates:
        - Growth
        - Maturation

          Of ovarian follicles
    - Builds up
        - Uterus Endometrial lining
            - During menstrual cycle
    - Promotes
        - Female secondary sex characteristics
    - Regulates ovarian cycle

Progestrone:

- Made by
    - Corpus luteum
        - Ruptured follicle after ovulation
- Functions:
    - Maintains endometrial lining
        - For implantation of zygote
    - Supports
        - early stages of pregnancy
    - Stops more ovulation
        - During pregnancy
    - Prepares breasts for
        - Lactation
            - By stimulating milk secreting tissues
    - Trigger menstruation

    - Drop in progesterone
        - If no fertilisation

Ovarian Cycle:

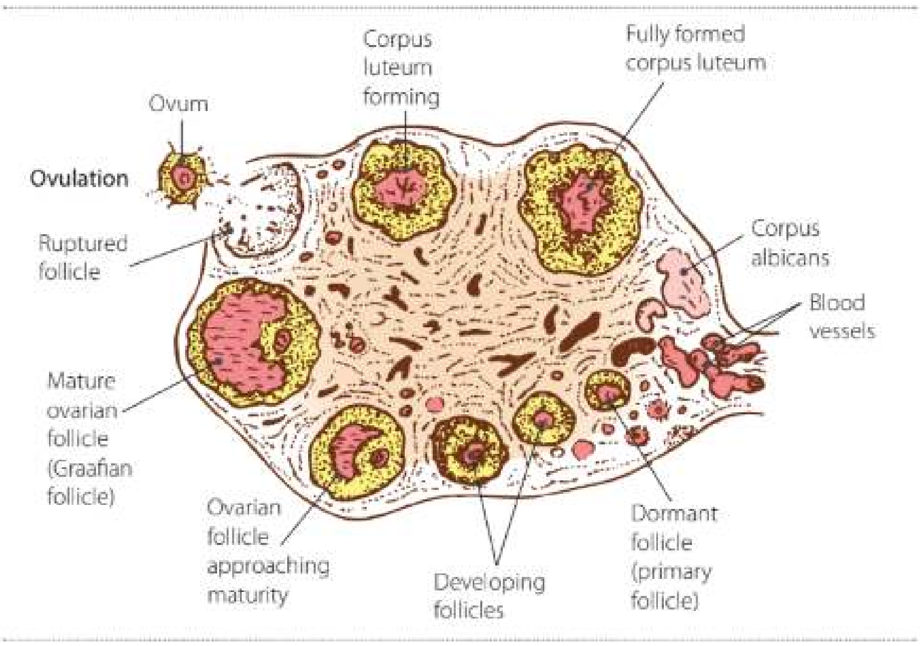
- ovaries
- Females born with all ova (eggs)
    - Eggs remain immature
        - Until puberty
- Ovarian + menstrual cycles
    - Start at puberty (menarche)
        - Continue → menopause
- Each cycle
    - Few follicles develop
        - 1 matures fully
- Normal cycle
    - 28 days
    - May vary

Follicular Phase:

- Follicle cells
    - Secrete Fluid
        - Push ovum to 1 side
- Dominant follicle → graafian follicle
- 10-14 days from primary → graafian follicle
- Follicles
    - Secrete oestrogen
        - → surge in LH
- LH surge
    - Trigger ovulation
        - Release egg
    - Preps for luteinising phase
- Ovulation:
    - Graafian follicle bursts
        - Release egg
    - Egg → uterine tube
        - By cilia
    - Fertilisation if

        - Sperm there
    - Ovum survives
        - 12-24 hrs after ovulation

- **FIGURE 2.34** Diagrammatic representation of the ovarian cycle, including ovulation



Luteal Phase:

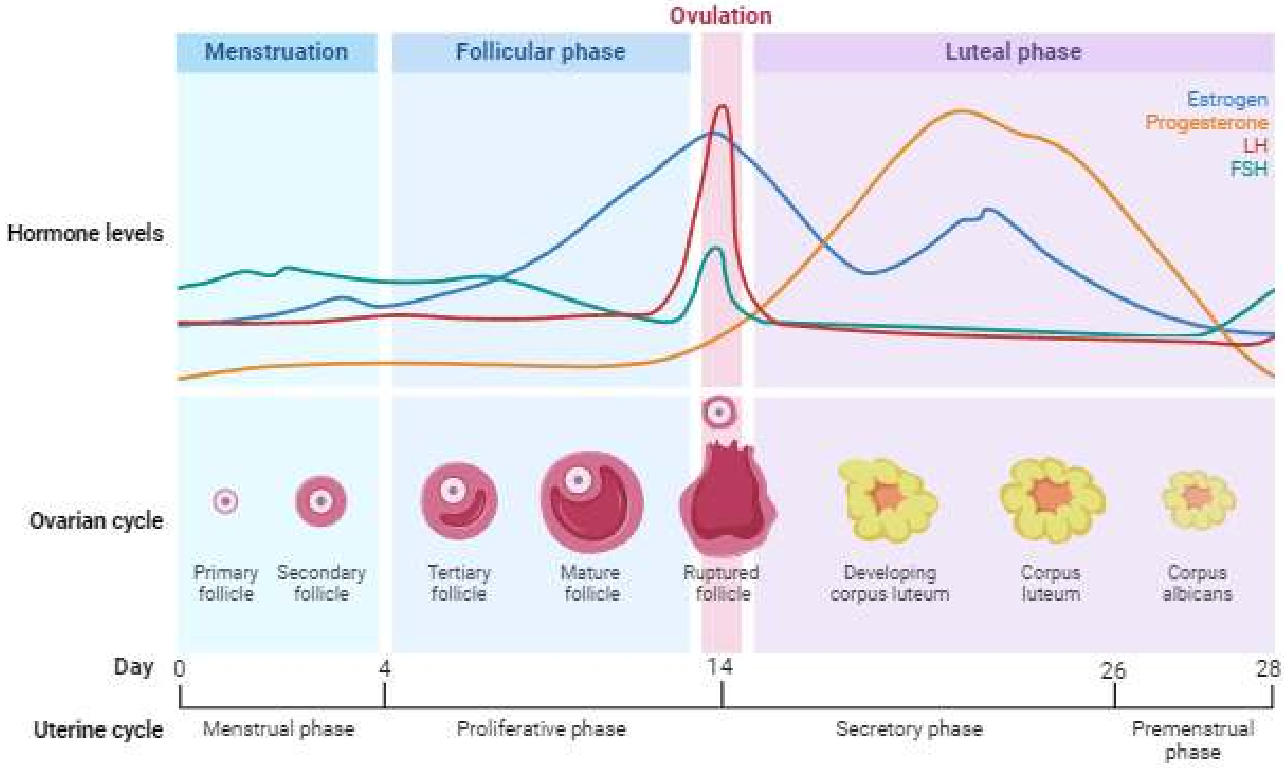
- Last 14 days
- Starts after ovulation
- Burst follicle → corpus luteum
- Corpus luteum
    - Secretes progesterone
        - Prepares uterus for pregnancy
        - Supports endometrial lining
        - If fertilisation
            - Stops menstruation
- If no fertilisation
    - Corpus luteum degenerates
    - Progesterone levels drop
    - Menstruation begins

Menstrual Cycle:

- Uterus
- ~28 days
- Stages:
    - Follicular

        - 1-13 days
    - Ovulation
        - 14th day
    - Luteal
        - 15-28 days
- Menses
    - 1-4 days
    - Breakdown, shedding of uterus lining (endometrium)
- Follicular phase:
    - FSH rises
        - Stimulates follicle development
        - Oestrogen increase
            - By follicles
        - Endometrial
            - Repair
            - Thickening
- Ovulation:
    - Day 14
    - peak in LH
    - Mature egg released
        - From follicle in ovary
- Luteal phase:
    - Corpus luteum made
        - Secrete progesterone
            - Maintain endometrial lining
- If ❌ fertilisation:
    - Hormones drop
        - → breakdown of endometrium
            - menstruation
        - Start of new cycle

- FSH → follicles develop → Oestrogen → LH → mature follicle → Peak in LH → egg release → corpus luteum → Progesterone → maintain uterus lining → hormone drop (not fertilised) → menses
- Hypothalamus → gnrh → pituitary → fsh → rest of cycle



Reprinted from "Ovarian Hormones Throughout the Menstrual Cycle", by Rosalba Lopez at BioRender.com (2024). Retrieved from https://app.biorender.com/biorender-templates

▶ Ovulation - Nucleus Health

- If ✅ fertilisation:
  - Fertilised ovum
    - Implants in uterus
  - Pregnancy
  - Uterine wall maintained by
    - Progesterone
    - Oestrogen
  - Hormones o and p
    - First made by corpus luteum
    - Then by placenta
  - Placenta forms
    - Attach embryo
      - To uterine wall
  - Placenta secretes
    - Progestrone

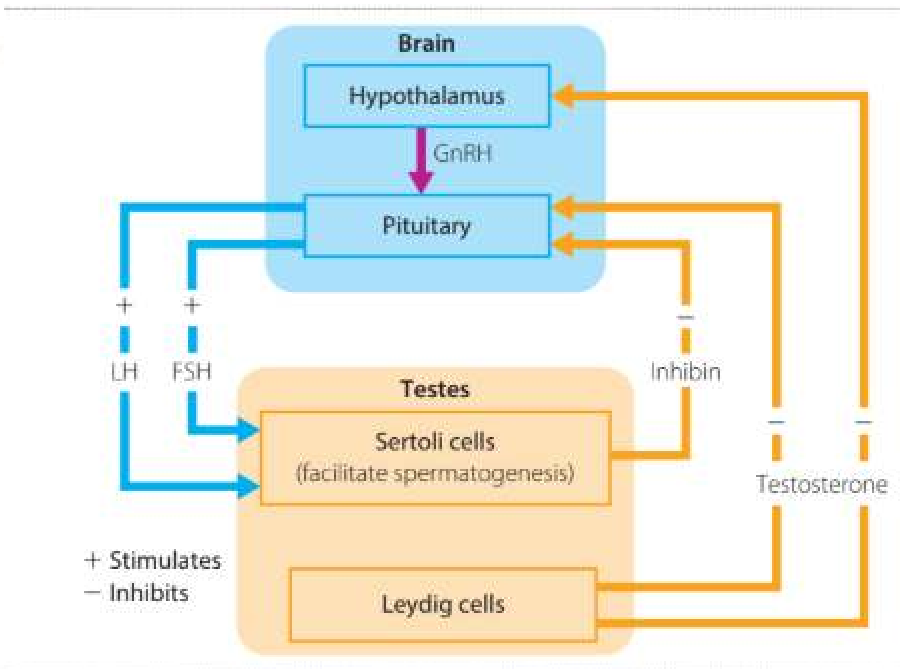
    - Oestrogen
    - Human chorionic gonadotropin (HCG)
        - Maintains corpus luteum, keep secreting o + p

          To maintain pregnancy
- When placenta able to secrete hormones
    - Corpus luteum degenerates

Male Reproductive Cycle - Hormones:

- Sperm production
    - Hypothalamus
    - Pituitary gland
    - Leydig cells in testicles
- LH:
    - Production of testosterone
- FSH:
    - Production of protein by
        - sertoli cells in testes
            - To maintain testosterone level
                - High enough for spermatogenesis
- Basically
    - LH + FSH → spermatogenesis + testosterone
        - In testes
- Inhibin:
    - Reduce FSH

FIGURE 2.36 Hormonal control of sperm production: LH and FSH from the pituitary stimulate spermatogenesis and testosterone secretion by the testes. Testosterone inhibits the secretion of GnRH by the hypothalamus, and testosterone and inhibin from the testes inhibit the secretion of LH and FSH by the pituitary.



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Fertilisation:

- Haploid egg + sperm nucleus
    - Fuse
        - → diploid nucleus zygote
- Zygote divides
    - As travels along fallopian tube
        - Into uterus
    - Developing into embryo

Pregnancy and Birth - Hormones:

- Pregnancy begin when
    - Embryo implants into uterine wall
- When embryo implants
    - Corpus luteum in ovary
        - Grows
        - Secrete hormones
            - For first 3 months of pregnancy
- Next 6 months
    - Corpus luteum shrinks
        - Degenerates slowly
        - Still present at childbirth
- Placenta produces hormones
    - After corpus luteum
    - To maintain pregnancy
- Oestrogen:
    - Promotes growth of endometrium lining
- Progesterone
    - Suppresses uterine activity
        - Supports fetal development
    - Reduce risk of fetus
        - Disturbed
        - Expelled by uterine contractions
- Human Chorionic hormone (HCG)
    - Maintains corpus luteum
        - For first trimester
- Placenta:

- Large disc of tissue
  - Made of maternal + fetal tissue
- Embedded in uterine wall
- Connected to fetus by
  - Umbilical cord
    - Carries blood vessels to and from baby
- Function:
  - Transport oxygen + nutrients
    - From mom → baby
  - Remove wastes
    - Carbon dioxide
    - Urea
    - From baby → mom for excretion

Birth:

- Must happen for birth:
  - Muscles in uterus contract
    - To expel baby
  - Tissue of cervix
    - Must soften so
      - Cervix can dilate
        - For passage of baby
- Prostaglandins
  - secreted by wall of uterus
    - Initiate labour
  - Make uterus tissue more sensitive to
    - Oxytocin
- Oxytocin:
  - Promotes
    - coordinated contraction
      - Of uterus smooth muscle
    - Softening of cervix
      - So baby born
- oestrogen + progesterone
  - Decline during labour
- Contractions become stronger
- Intensity of contractions are greater

        - Beta endorphins
            - Natural form of pain relief
            - Feelings of great happiness
- Adrenaline:
    - Very strong contractions
    - Lead to birth
- Continued oxytocin release
    - Expel placenta (afterbirth)
    - Reduce risk of bleeding
- Prolactin:
    - Made by
        - Pituitary gland
        - Sometimes uterus, breasts
    - Stimulate milk production
    - + oxytocin
        - → maternal care, bonding

Impact of Science on Agricultural Reproduction:

- "Evaluate" meaning in dot point:
    - List components
    - Describe characteristics and features of
    - Explain purpose and function
    - Provide examples
    - Determine effects and impacts
    - Make a judgement
        - Determine strength of impact
- Manipulation of plant and animal reproduction in agriculture include:
    - artificial insemination
    - selective breeding
    - selective cross-breeding
    - genetic engineering

Cell Replication:

Cell Division:

Mitosis:

- Eukaryotic cells
- Multicellular organisms
- 1 single cell divides
  - Into 2 daughter cells
    - Have identical 46 chromosomes each
- Is division of nucleus
- Chromatin:
  - Dna wound around protein
- Genes:
  - Units of heredity
  - Code for inherited characteristics of an organism
- Sister chromatids:
  - 2 chromosome copies
  - In x shape
  - Have centromere holding them together

Stages of Mitosis:

- Interphase:
  - Dna is copied
    - 2 identical full sets of 46 chromosomes
  - Dna:
    - known as chromatin
    - diffused, spread out
    - begins to separate into chromosomes
  - Outside of nucleus:
    - centrosomes:
      - Each have pair of centrioles
        - critical for cell division
  - Microtubules extend from centrosomes
- Prophase:
  - Dna:
    - shortens + thickens by coiling

        - separates into chromosomes (x shape)
    - Each chromosome
        - has 2 copies of the dna
    - Each copy (each x)
        - is sister chromatid
        - joined by 1 centromere
            - (middle connection part)
    - Nucleus membrane breaks down
        - is then not visible
    - Spindle forms across cell
        - using broken Nucleus membrane
            - (looks like lines of lines of longitude)
- Metaphase:
    - Middle → META
    - Chromosomes
        - line up across centre of cell
    - Membrane spindle
        - attaches to centromeres of chromatids
- Anaphase:
    - Spindle fibres contract
    - Sister chromatids pulled by
        - their centromeres
            - to opposite ends of cell
    - Sister chromatids (x chromosomes)
        - separate
    - Each single chromatid
        - becomes separate chromosome
    - Daughter chromosomes:
        - separated sister chromatids
- Telophase:
    - Chromosomes gather
        - at opposite poles of cell
    - Spindle breaks down
    - nucleus + nucleolus membranes
        - are made around daughter nuclei chromosomes
    - Mitosis finished
- Cytokinesis:

- Splitting of cytoplasm to
    - make 2 separate daughter cells
- Animal cells:
    - Cytoplasm:
        - tightens around center of cell
            - between the 2 daughter nuclei
        - Pinches off
            - creates 2 daughter cells

Role and importance:

- Growth
    - Of multicellular organisms
- Repair
    - Of damaged tissue
    - Replacement of worn out cells
- Asexual reproduction:
    - Growth of plants from cuttings
    - Organism cloning
- Genetic stability:
    - Precise,
    - Equal distribution of chromosomes
        - To each daughter cell
            - So all resulting cells
                - Have same chromosome number
                - Same genetic information
                    - As each other + parent cell

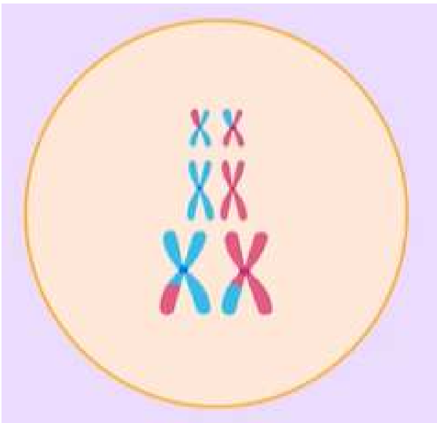
Meiosis:

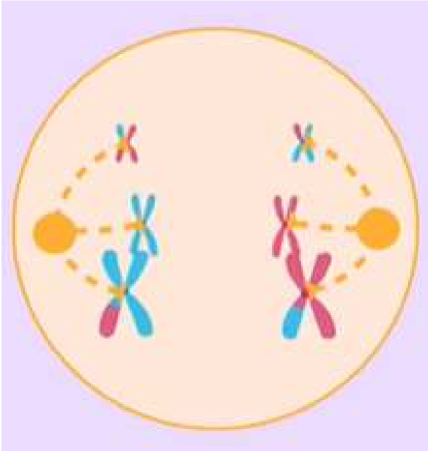
- In sexual reproductive organs
    - of plant, animal
- Result:
    - sex cells
        - (gametes)
- 2 stages:
    - meiosis I
    - meiosis II
- 1 diploid cell:

- → 4 haploid gametes

Stages of Meiosis:

Meiosis I:

- Interphase I:
    - Same as mitosis
- Prophase I:
    - Same as mitosis but
        - Chromosomes line up in homologous pairs
    - Order of chromosome lineup
        - may be random
            - random assortment
    - Chromosomes exchange genes with each other
            - Crossing Over
- 
- Metaphase I:
    - Same as mitosis
    - Chromosome pairs line up
        - along centre of cell
    - Spindle fibres attach to
        - centromeres of chromosomes
- Anaphase I:
    - Same as mitosis but
    - Centromeres not broken
    - Sister chromatids not separated
    - Half of 2 sets of chromosomes
        - move to each end of cell
        - (1 set moves to either side of cell)

- 
- Telophase I:
    - Same as mitosis
- Cytokinesis I:
    - Same as mitosis

Meiosis II:

- Same as mitosis
    - Sister chromatids broken in each of 2 cells
        - From meiosis 1
            - Making 4 cells
- Each daughter cell from Meiosis I
    - has 23 pairs of chromosomes
    - (total 46 chromosomes)
- After meiosis II:
    - each of 4 grand daughter cells
        - have one pair of 23 chromosomes
        - (total 23 chromosomes)
- Process same as normal mitosis but
    - → 4 grand daughter cells with:
        - Half amount of single chromosomes
            - in each sex cell (23)
                - than with body cell (46)
        - different chromosomes, genes
            - than original parent cell
            - Than other granddaughter cells

Role and importance:

- Prevents

        - number of chromosomes
            - in each successive generation
                - from doubling
- Ensures:
    - that half of total chromosomes from 1 parent
        - is passed onto offspring
    - chromosome number in species
        - is maintained in sexual reproduction
- Adds genetic variation in species
    - Crossing over
    - Random assortment

Why DNA needs to be replicated exactly:

- So each generation of cells
    - has same genetic code
- Genes passed from parent cell → offspring
    - Maintains genetic stability of species
- function in:
    - Controlled
    - coordinated way
- proteins, enzymes
    - Used in all parts of body, cells
    - Majority of cells
    - Used in biochemical reactions (enzymes)
- So proteins, enzymes
    - Are made correctly
    - carry out processes correctly
    - Without error
- Change or error in dna
    - May change end products of cell
    - Harmful effect on cell
        - Interfering with structure functioning

Cell Cycle:

- Cell division + enlargement
    - in repetitive sequence
- 4 phases:

- G1:
    - gap phase for cell growth
    - Cell enlargement
        - content of cell (minus dna) is doubled
    - metabolic changes prepare cell for division
- S:
    - Synthesis
    - dna replicates in cell
- G2:
    - gap phase 2
    - enzymes check errors in
        - replicated dna
            - repair them
- M:
    - Mitosis
    - division of cell nucleus into 2
    - + cytokinesis
        - division of cytoplasm into 2
            - For both nuclei
- G0:
    - cell cycle stops

- actively dividing tissue
    - C → G1 to restart cell cycle
- G1 → G2 = interphase

DNA Structure - Watson and Crick Model:

History of Discovery:

- Friedrich Miescher 1869:
    - identified DNA as a distinct molecule
- William Johannsen 1909:
    - discovered genes carried in chromosomes
- 1910:
    - chromosomes discovered
        - + they were made of DNA and protein
- Oswald Avery 1944:
    - experimental proof using bacteria to show DNA

        - as hereditary material and not protein
- Erwin Chargaff 1951:
    - proposed chargaff's rule
- Linus Pauling 1953:
    - in america
        - proposed DNA as hereditary molecule
            - its structure as a triple helix
- Watson and Crick 1953:
    - Used chemical + x ray evidence
        - to model molecular structure of DNA
    - Model shows that dna was:
        - double stranded
        - had paired nitrogenous bases
            - held by hydrogen bonds
        - twisted into a spiral ladder
        - storing large amounts of genetic information
            - in order of bases
            - that make up middle of dna structure
    - Model:
        - was called “double helix”
        - could be used to explain how dna could self replicate
    - Maurice Wilkins showed Watson
        - Rosalind Franklin's
            - dna x ray crystallography photograph
        - (without first asking her permission)
    - They used
        - crystallography + chemistry data
            - to solidify their models argument
                - won nobel prize 1962

DNA Structure:

- Antiparallel
    - Stands run parallel
    - In opposite directions
- Set distance between ‘backbones’ of dna
- Backbone made of
    - Deoxyribose sugar

- Phosphate
- Each base
  - 1 end attached to backbone
  - 1 end hydrogen bonded
    - To complementary base
- Bases line up
  - Between backbones
- has 3’ and 5’ end
- Length of dna = no. of base pairs
- Sugar + phosphate
  - Alternate in backbone

Nucleotide Pairing and Bonding:

- Purine base with pyrimidine base
- Purine bases:
  - Adenine
  - Guanine
- Pyrimidine bases:
  - Thymine
  - Cytosine
- Adenine with Thymine
  - A with T
- Guanine with Cytosine
  - G with C
- To remember base pairings:
  - Apples in the Tree
  - Cars in the Garage
- Dna strands
  - Held together by
    - Weak hydrogen bonds
- Double bond:
  - A with T
- Triple bond:
  - C with G
- Chargaffs Rule:
  - Ratio of complementary base pairings:
    - Always 1 : 1

- A : T → 1 : 1
- C : G → 1 : 1

DNA as Template for Replication:

- Watson crick model
    - Showed all requirements for
        - Hereditary material
- Requirements:
    - Dna can carry in coded form
    - All instructions for
        - Formation
        - Function of cells
            - Despite it having only 4 nitrogenous bases
    - Structure allows
        - For own replication
        - To copy
            - Each strand can act as template
            - For enzymes to make
                - New complementary strand
    - Dna transferred
        - From 1 gen → next gen
        - In form of
            - Chromosomes
                - Carried by gametes

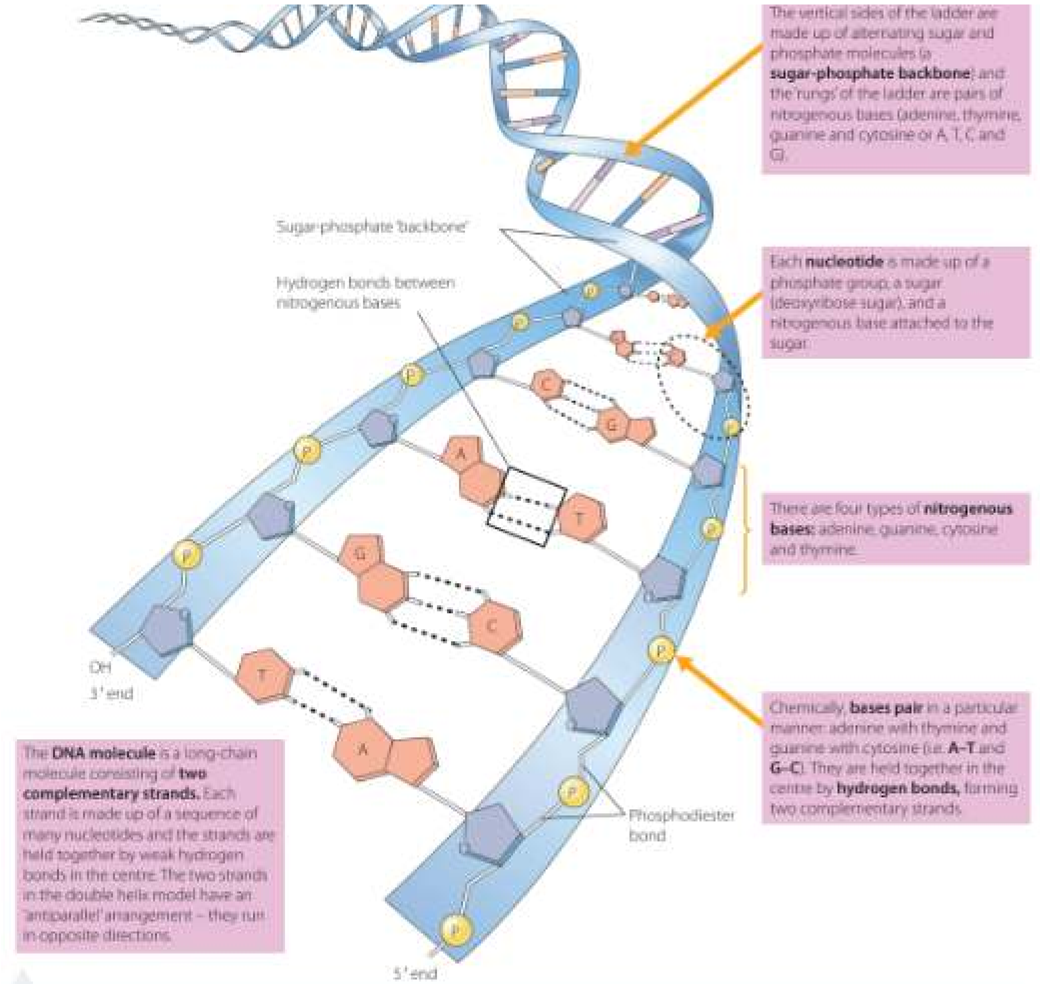


FIGURE 3.12 An annotated diagram of the DNA double helix molecule, showing sugar-phosphate backbone and base pairing

TABLE 3.3 Validation of the model: Watson and Crick's findings, and the evidence on which they were based

FINDING	EVIDENCE-BASED REASONING
The whole DNA 'ladder' molecule, instead of being flat, spirals and is therefore known as the 'double helix'.	*Evidence*: X-ray crystallography suggested a helix measuring 3.4 nm for every turn, and this fitted the model where exactly 10 base pairs would measure 3.4 nm in length and make up one twist of the helix.
The four types of nitrogenous bases: adenine (A), guanine (G), cytosine (C) and thymine (T) always pair as A–T and C–G.	*Evidence*: Chargaff's rule stated that DNA from any cell has a 1:1 ratio of pyrimidine and purine bases and, more specifically, that the amount of guanine is equal to cytosine and the amount of adenine is equal to thymine. Watson and Crick found complementary base pairing of pyrimidine–purine (that is, A–T and G–C) was critical in keeping the backbones of DNA equidistant.
The two complementary strands of DNA are held together in the centre by hydrogen bonds that form between the complementary bases.	*Reasoning*: Chemically, when A bonds to T, a double hydrogen bond may form and when G bonds to C, a triple hydrogen bond may form.
The strands of the backbone must be identical and run in opposite directions (antiparallel).	*Evidence*: DNA crystal images, generated by Rosalind Franklin, looked the same when they were turned upside down or backwards. *Reasoning*: The backbones are made of sugar-phosphate molecules, based on the ratio of these in the chemical analyses.
Each DNA strand serves as a template for the production of a complementary strand, allowing the double-stranded molecule to self-replicate.	*Reasoning*: The two complementary strands of DNA could 'unzip' or open up if the hydrogen bonds break between the base pairs, allowing them to replicate.
Watson and Crick could form hypotheses about DNA's structure by building physical models of how the atoms fit together. Sometimes they used cardboard cut-outs to represent the four nitrogenous bases and other subunits of DNA. They played with these on a desk, like a jigsaw puzzle. They initially made an error in the configuration of the rings in thymine and guanine. By changing the arrangement of atoms and making new cut-outs, based on a suggestion from Jerry Donahue, an American scientist, they found the perfect fit: complementary base pairing, ratios to reflect Chargaff's rule and the hydrogen bonding of purines to pyrimidine bases.	

-

Nucleic Acids:

- DNA, RNA

- Are polymers
    - Made of monomers
        - Called nucleotides
- Each nucleotide:
    - 3 parts:
        - 1 Deoxyribose Sugar (DNA), Ribose sugar (RNA)
        - 1 phosphate
        - 1 nitrogenous base
    - Nucleotide attached to sugar
- Genetic code in
    - Continuous sequence of bases
- Base sequences differ
    - Making genetic code

DNA instructs Formation of Proteins:

- Order of 4 bases
    - Determine sequence of amino acids
        - In protein

DNA Replication - Watson and Crick Model:

- Creation of
    - 2 identical DNA double helix molecules
        - from 1 original DNA molecule
- Dna replicates
    - during interphase in cells
- Semi-conservative process:
    - uses one strand from old dna
        - as template to create new strand
    - Increases accuracy
        - in copying base pairs exactly
- Features:
    - Dna unwound from spiral configuration
        - Before strands separated
    - Lots of enzymes needed
    - 2 strands of dna
        - Run in opposite directions
    - Nucleotides added in both directions

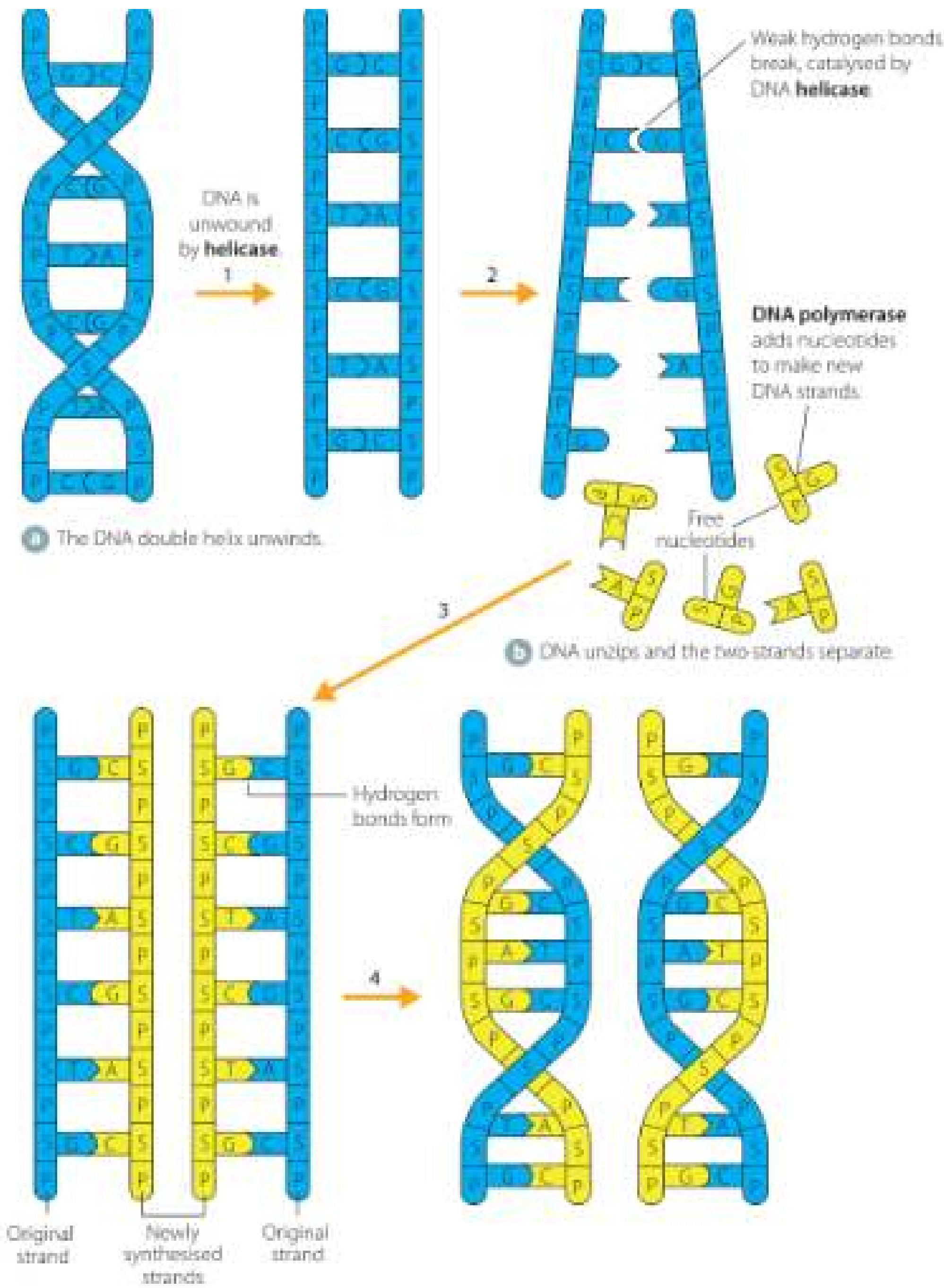
        - Continuous on 3' end
        - Fragmented on 5' end
    - Correction of replication errors

Process:

1. Unwind DNA double helix:
    - Helicase enzyme:
        - unwinds twisted helix of dna
        - Separates the 2 dna strands
2. Separate 2 dna strands:
    - Helicase
        - breaks hydrogen bonds
            - between the complementary nucleotides
        - uses ATP as energy
    - Dna strands separate
        - expose nucleotide bases
    - Starts down bottom
    - Replication fork:
        - made where dna is still joined
    - Single stranded binding proteins (SSBs)
        - bind to
        - + stabilize separated single strand DNA
3. Nucleotides added against each single strand:
    - Each strand of dna molecule
        - Is template for new strand of dna
    - When synthesis starts
        - RNA primer attaches to DNA
            - Primer Made by enzyme primase
    - Enzyme DNA polymerase III
        - adds nucleotides to strands
    - Nucleotides
        - floating free in liquid of nucleus
        - added with complementary base partner
    - Keep pairing
        - A to T
        - G to C
    - Dna polymerase

        - binds nucleotides together
            - along where backbone is to be
            - forms sugar-phosphate backbone
                - on newly made strand
    - Nucleotides added from
        - 3’ side of dna
    - Adding nucleotides is antiparallel
    - Leading strand (3’ from bottom):
        - nucleotides added continuously
            - in same direction as replication fork opens up
                - (bottom of dna → replication fork)
    - Lagging strand (5’ from bottom):
        - nucleotides added backwards
            - starting from replication fork → bottom
        - Nucleotides added in chunks
            - Okazaki fragments
        - Replication is discontinuous
        - Okazaki fragments
            - joined by enzyme ligase
                - forms continuous strand
4. Replication errors are identified and corrected:
    - Dna polymerase I
        - backtrack + edit dna strands
        - removes the incorrect base inserted into new strand
            - replaces it with correct base
    - enzyme ligase
        - Joins 2 strands together
        - Joins Okazaki fragments togeth.
    - Another dna polymerase enzyme
        - check final base pairing
            - Recognise base mismatch error
                - performs repairs
    - Each newly made dna molecule
        - has
            - one strand from old dna
            - one newly made one
    - Dna rewinded

- into double helix
- Final result:
    - 2 double helix dna molecules
        - identical to each other + original molecule



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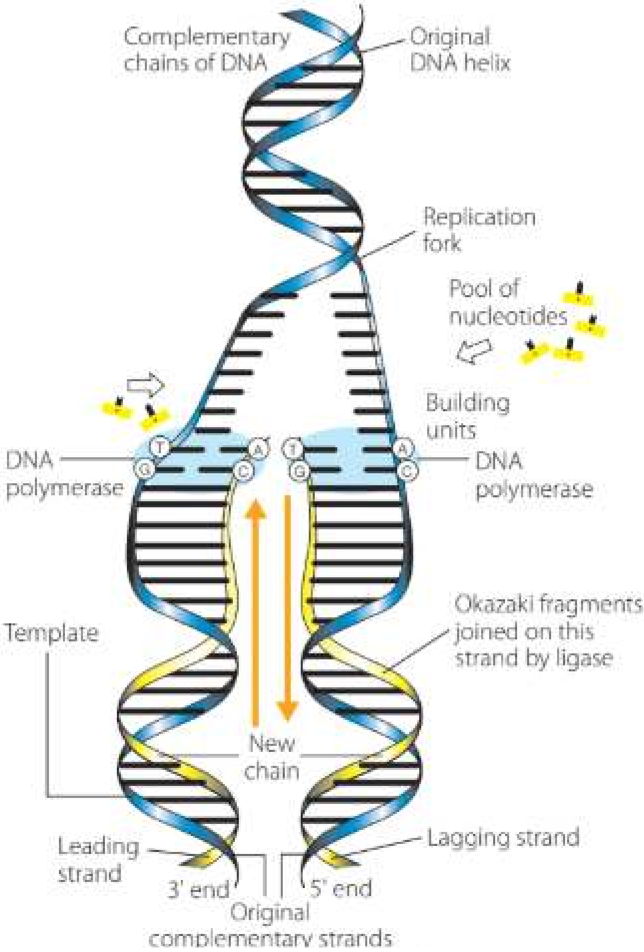

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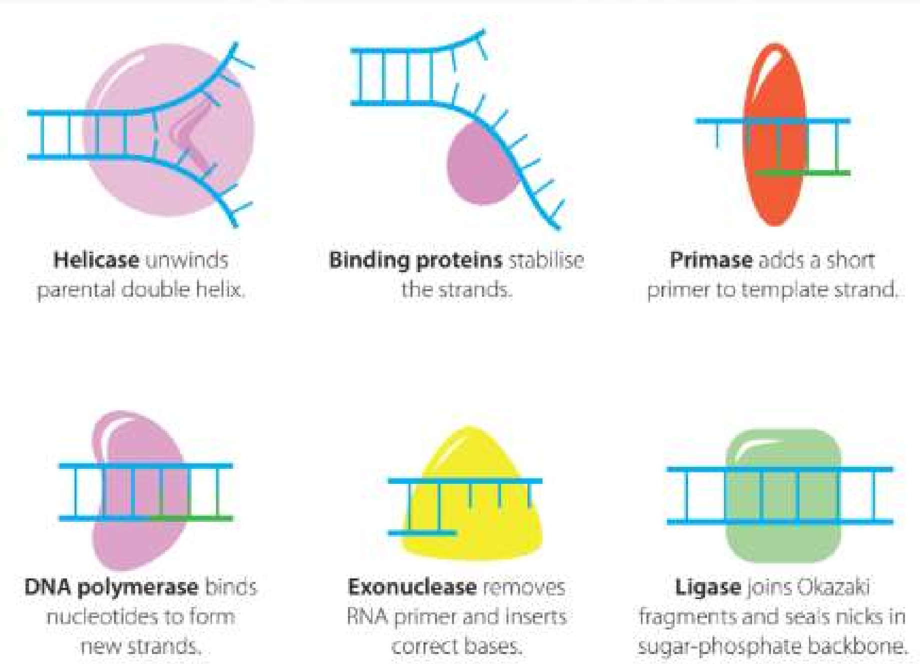
Enzymes:

TABLE 3.4 Enzymes that regulate DNA replication

ENZYME	FUNCTION IN DNA REPLICATION
Topoisomerase (e.g. gyrase)	• Relaxes DNA from its supercoiled state, always working ahead of the replication fork
Helicase	• Follows topoisomerase and unwinds the double helix by breaking hydrogen bonds between bases, causing the two strands to separate and creating a replication fork
Primase	• Connects RNA primer to a strand to initiate DNA replication • Synthesises a short complementary RNA molecule, which binds to DNA, serving as the starting point for DNA synthesis by polymerase
DNA polymerase III	• Synthesises new DNA strands, using existing strands as a template. Nucleotides are added to a growing strand from the 3′ end • Joins the phosphate group of each nucleotide to the previous one, creating phosphodiester bonds between these molecules to make the sugar-phosphate backbone
DNA polymerase I	• Mainly functions in 'editing' – recognises and repairs base pairing errors (exonuclease) • Also has a function in replication (removing primers ahead of the main polymerising enzyme)
DNA polymerase II	• Editing function, but no exonuclease activity
Ligase	• Connects and seals the two strands of the DNA molecule and also connects Okazaki fragments

- On the leading strand – DNA moves along in the same direction as the developing replication fork, and nucleotides are added in one long chain.
- On the lagging strand, DNA is added one 'chunk' at a time – called Okazaki fragments (about 100–150 nucleotides long) and these are then joined up.

FIGURE 3.16
Enzymes involved in DNA replication



Importance of Accuracy during DNA Replication:

- Heredity:
  - Inheritance of genes
  - Genetic material from
    - Cell → cell
    - Gen → gen

    Needs to be accurate
- Gene expression:
  - Protein synthesis
  - Genetic instructions given to cell to
    - create structure
    - Ensure functioning

    Must be accurate
- Errors in genes controlling cell cycle
  - → changes in cell division
  - Cell death
    - → cancer

The Continuity of Species:

- Continuity of species:
  - continuous survival of species
    - resulting from characteristics from genes

- being passed from parents → offspring
- Relies on
  - passing of consistently accurate genetic information
    - + some variation to
      - allow for species to
        - adapt + survive
          - in changing environment
- Accurate dna replication → genetic stability:
  - for survival of individual
- Gene mutations → genetic variation:
  - for evolution + survival of species
    - in changing conditions
- Genetic Continuity:
  - Dependent on:
    - daughter cells of mitosis to
      - have same number + make up of genes
        - as original cell
    - offspring of sexually reproducing organisms to
      - have same number of genes
        - as the parent organisms
          - + not have detrimental or lethal gene variations
  - Ensures new cells, organisms
    - have all genes they need, functioning
      - to survive
- Species Continuity:
  - Desirable traits + some random errors (maybe from gene mutation)
    - passed on
  - Allows species to survive
    - if environmental change occurs
- Natural selection:
  - individuals in population best suited to environment
    - survive + reproduce
      - passing on advantageous genes
        - to offspring
  - removes individuals
    - with damaged genes
      - for increased species continuity

- Mixing of parent genes
    - during sexual reproduction
    - + genes from mutation
        - → genetic diversity
- Negative effects:
    - Errors in dna replication, cell division
    - + reduced ability of dna repairing
        - → threatened continuity of species
            - as some mutated genes or change in gene numbers or types
                - may lead to disease or death
- Mitosis + Meiosis:
    - Mitosis:
        - Equal distribution of chromosomes
            - when forming body cells
                - → genetic continuity
        - Growth + repair of body cells
        - creation of organs for organism functioning
            - → growth + functioning until reproduction age
                - → continuity of species
    - Meiosis:
        - Equal distribution of chromosomes
            - when forming gametes
        - crossing over + random assortment
            - → mixing of parental genes during sexual reproduction
                - → genetic diversity
        - chromosome number in species
            - maintained during sexual reproduction
                - prevents doubling across generations
- DNA replication:
    - Consistent dna replication
        - before mitosis meiosis
            - ensure species
                - genetic stability
                - + continuity
- Genetic continuity ensured by:
    - correct dna repair
        - by enzymes

        - during dna replication
- Mutation
    - changes in DNA from mutation (spontaneous or mutagen-induced)
    - can be good or bad depending on environment or specific mutation

DNA and Polypeptide Synthesis:

Prokaryotic DNA:

Location and Structure:

- Prokaryotic cells
    - Simpler than eukaryotic cells
- Have
    - Single
    - Circular

      Chromosome
- Chromosome
    - Has No membrane around it
    - Floats in cytoplasm
- DNA
    - In dense region
        - Nucleoid
    - Codes for proteins
        - Made on ribosomes
            - In cytoplasm
    - Double stranded
    - Circular
    - Twisted + coiled
        - For compact storage

Non-Chromosomal DNA:

- Prokaryotes may have
    - Small circular dna pieces
        - “Plasmids”
- Plasmids
    - Float freely in cytoplasm
    - Replicate independently

- Carry genes for
    - Non-essential but advantageous traits
        - E.g. antibiotic resistance

Packaging of Prokaryotic DNA:

- ~1300 nanom long
- Is tightly coiled
    - To fit in small cell
        - ~3 nanom
- Loops around
    - Central dense protein (scaffold)
        - To form nucleoid
- Dense protein
    - Different from histones in eu cells
- Prokaryotes Regulate gene expression
    - Using mechanisms
        - E.g. operon system

Eukaryotic DNA:

General Characteristics:

- In
    - Membrane bound nucleus
- Is linear
- Dna coils around histone proteins
    - To form nucleosomes
- Genomes have
    - Large amount of non-coding DNA (introns)
- Non-coding DNA
    - Does not directly code for
        - proteins
        - or RNA
- Coding DNA (3% in humans)
    - Is made of exons
        - Ex for 'expressed'
        - Code for
            - proteins

- or RNA

Non-nuclear DNA:

- Mitochondria + chloroplasts
  - Have their own DNA
- Inherited separately from nuclear dna
- Mitochondrial dna (non-nuclear DNA)
  - mtDNA
  - In respiratory organelles of cells

Eukaryotic DNA vs. Prokaryotic DNA:

-

Feature:	Prokaryotic DNA:	Eukaryotic DNA:
Location:	- In nucleoid - Not membrane bound	- Enclosed in - Membrane bound nucleus
DNA Shape and Structure:	- Single - Circular - double stranded DNA molecule - in double helix	- Multiple, linear - double stranded chromosomes - Chromosomes with linear structures

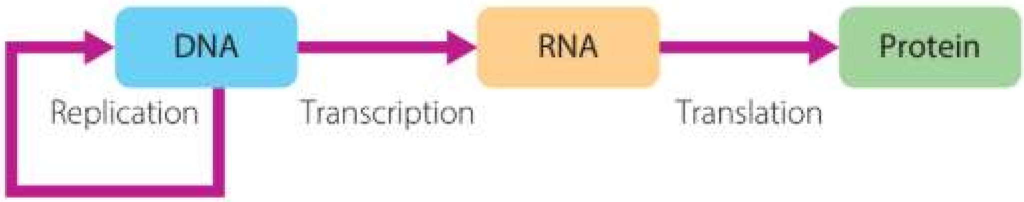
DNA packaging	- Supercoiled
	- looped around scaffold (non-histone) proteins
		- Tightly packaged around histone proteins
		  - Forms nucleosomes
		    - Further compacted → chromatin

Non-coding DNA	- Minimal - Or absent; - DNA mostly coding sequences	- Has large amounts of non coding DNA

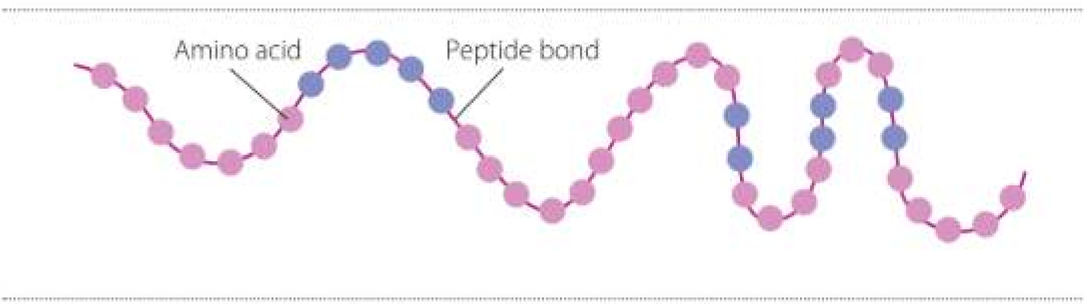
-

Polypeptide Synthesis:

- Francis crick
    - Proposed central dogma of molecular biology
        - Dna + rna = protein



-
- Dna codes for proteins
    - Through sequences
        - Codons (triplets of bases)
- Gene:
    - Section of dna that codes for
        - A specific protein
    - Determine order of amino acids
- Process involves:
    - Replication → dna
    - Transcription → (dna + rna)
    - Translation (rna + protein)
- Crick and brenner 1961
    - Showed that
        - Codons control
            - amino acid order
- Polypeptides:
    - Chains of amino acids
        - linked by peptide bonds
    - Can be up to 300 amino acids long



    -
- Proteins:

- Made when
    - One or more polypeptides
        - Fold into 3d shape

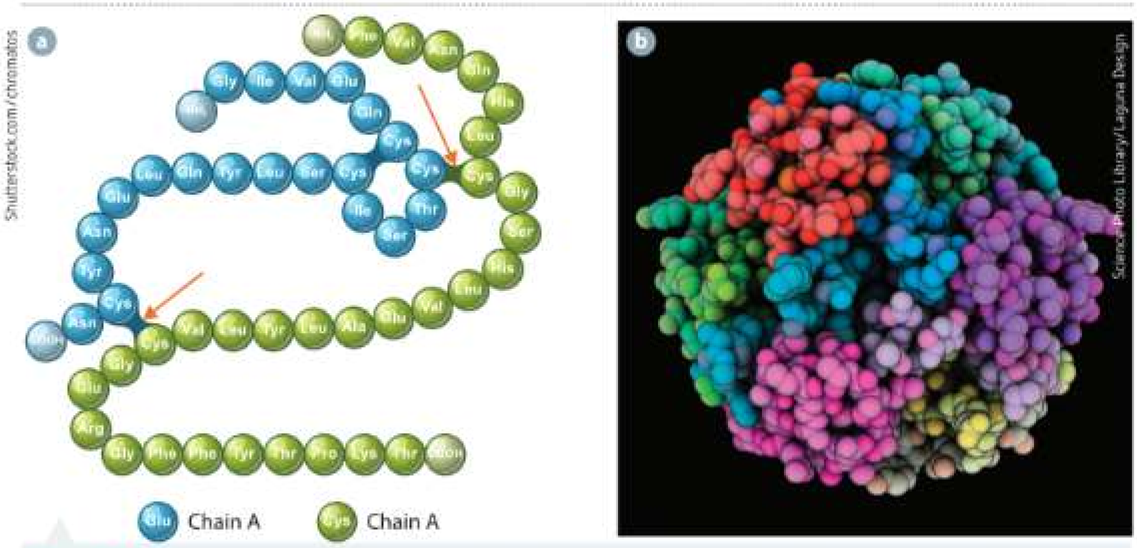


FIGURE 4.9 A protein, the hormone insulin: **a** a simple model showing that insulin is made up of two polypeptide chains joined at the points indicated (see red arrows); **b** a 3D molecular model of the structure of insulin

-
- 20 different amino acids
- Change in amino acid sequence
    - Alter protein
        - Shape
        - Function
- Gene concept:
    - When end products of gene
        - Made by cell
            - Gene is expressed
    - Certain genes expressed
        - In each cell type
    - Coded instructions for
        - Production of specific proteins
            - Are "switched on" in dna of that cell
        - Ensures each cell
            - Develop necessary structures
            - Adhering to type of tissue it is in
    - E.g.
        - Skin cells
            - Genes for production of proteins
                - melanin, keratin

- Switched on in cell

- Amino acids → polypeptides → chains of polypeptides by peptide bonds → proteins

Process of Polypeptide Synthesis:

- Dna stays in nucleus
  - Too large to pass through nuclear membrane
- Dna has full set of
  - Genetic instructions for cell
- Only needed instructions
  - Genes for specific proteins
    - Are accessed from dna
- Copy of instructions
  - Made as messenger rna (mRNA)
- mRNA
  - Carries transcribed (rna copied)
    - Genetic code from
      - Nucleus → cytoplasm in cell
- Ribosomes read mRNA
  - Translate into protein
- Protein
  - Is final product of
    - gene expression process
- Gene code needed → mRNA transcribed → nucleus → cytoplasm → ribosomes → translation into amino acids → proteins

Nucleic Acids in Polypeptide Synthesis:

DNA:

- Long chains of nucleotides
- Wound in double helix
- Sequence of nucleotide bases
  - Codes for sequence of
    - RNA nucleotides
      - → Sequences of amino acids
        - That form polypeptide chain

RNA:

- Made of nucleotides
- Single stranded
- Sugar:
    - Ribose
- Nitrogenous base:
    - Uracil
    - Instead of Thymine
- 3 types:
    - Messenger rna (mRNA)
    - Transfer rna (tRNA)
    - Ribosomal rna (rRNA)
- Mrna:
    - Single stranded
    - Not twisted into helix
    - In nucleus, cytoplasm
    - Carries information
        - From dna in nucleus
            - → ribosomes in cytoplasm
- Trna:
    - In cytoplasm
    - Cloverleaf shape
    - 75 nucleotides long
    - Has anticodon
        - That pairs with rna codons
    - Binds to SPECIFIC amino acid
        - Ensures correct sequence
            - during protein synthesis
- Rrna:
    - Forms part of ribosome structure
    - Synthesised in nucleolus of cell

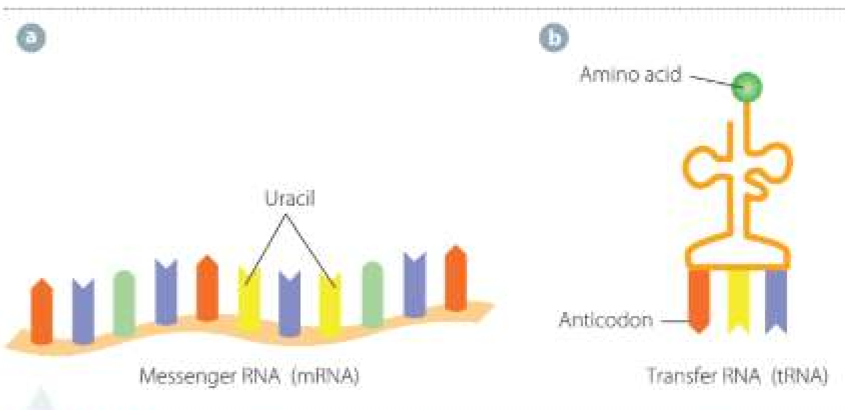


FIGURE 4.12 Three types of RNA: a mRNA, b tRNA and c rRNA (combined with protein to make

-

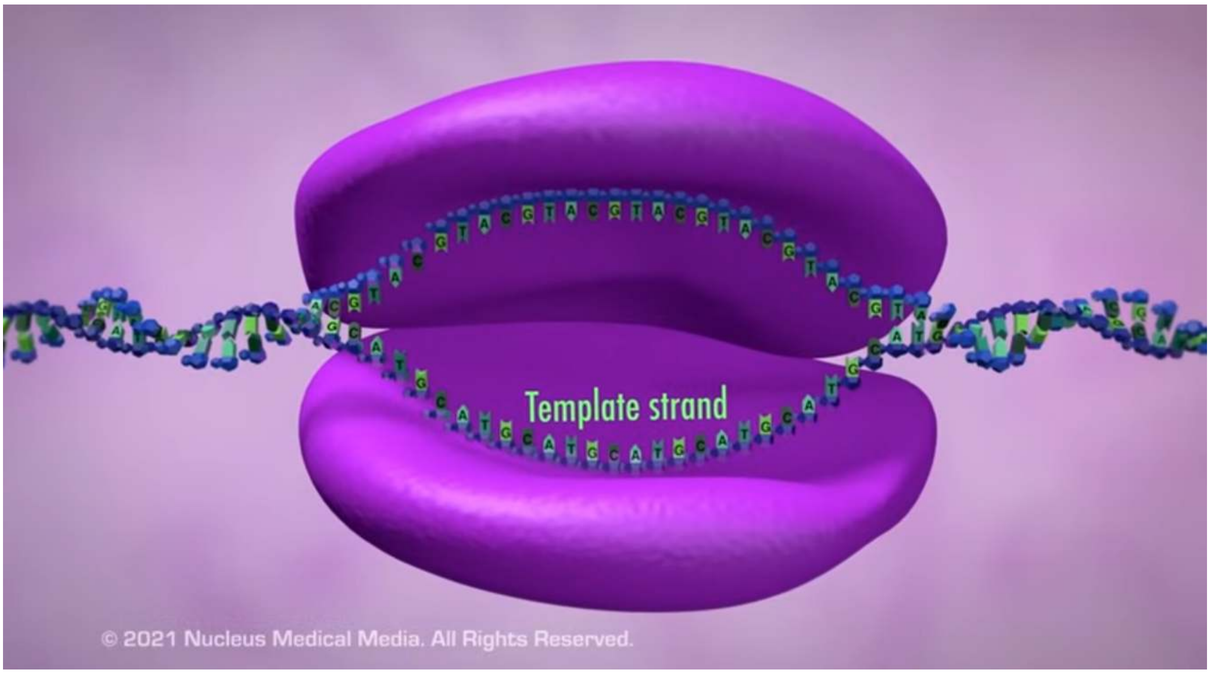
Transcription:

- RNA polymerase (enzyme)
    - Binds to section of DNA
    - Begins building
        - Chain of nucleotides
            - Form complementary strand of rna

Process:

1. Unzipping of gene to be copied

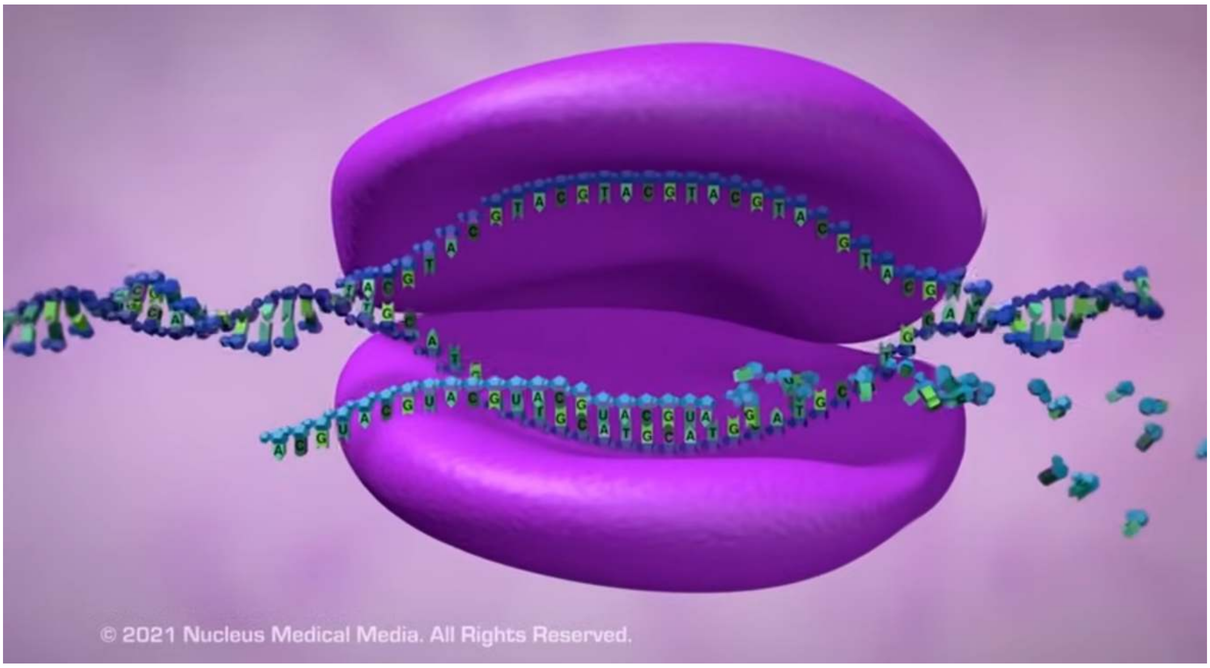
- Rna polymerase
- Binds to dna part
- Dna unzips
    - Dna unspirals
    - Hydrogen bonds
        - between 2 strands break
    - Strands separate over short length
    - Happens only in part of dna
        - that has gene to be used
- One strand of dna information
    - To make protein
    - "Non-coding strand"
    - What the rna polymerase using to make rna copy
- Other strand is
    - "Coding strand"
    - Has same code as mRNA being made
        - But uracil for thymine

- 

- Purple thing is rna polymerase

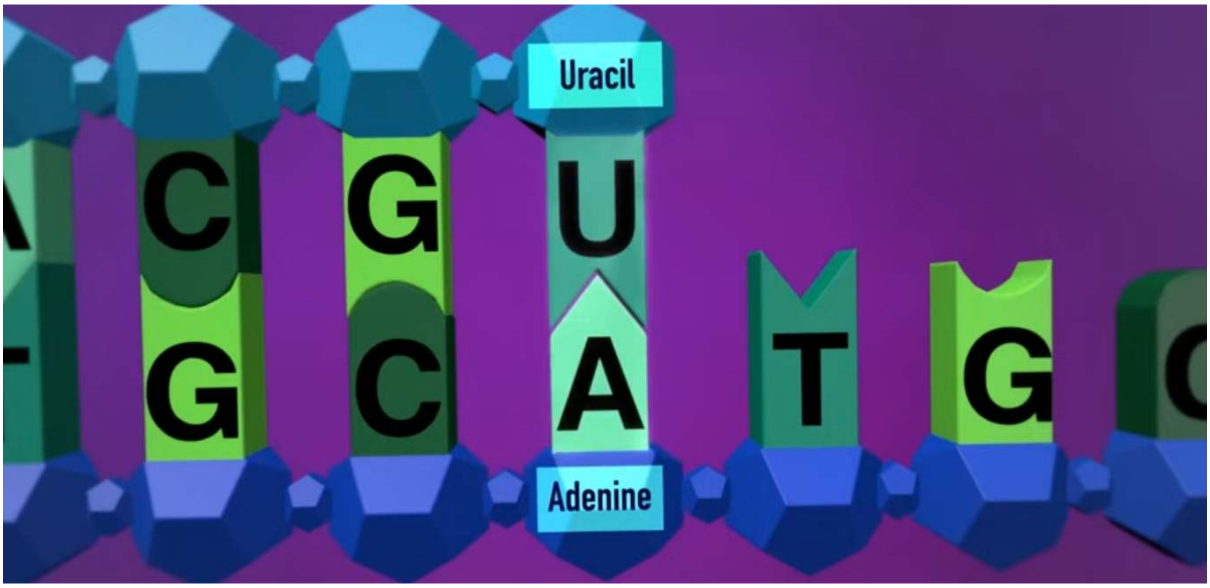
2. Transcribing Gene

- transcription of gene (copying of gene)
    - Controlled by RNA polymerase
- Non-coding strand
    - Acts as template
- RNA nucleotides assembled
    - Form complementary single stranded
        - mRNA molecule
        - (dna transcribed → mrna)

- 

- Sequence of nucleotide bases

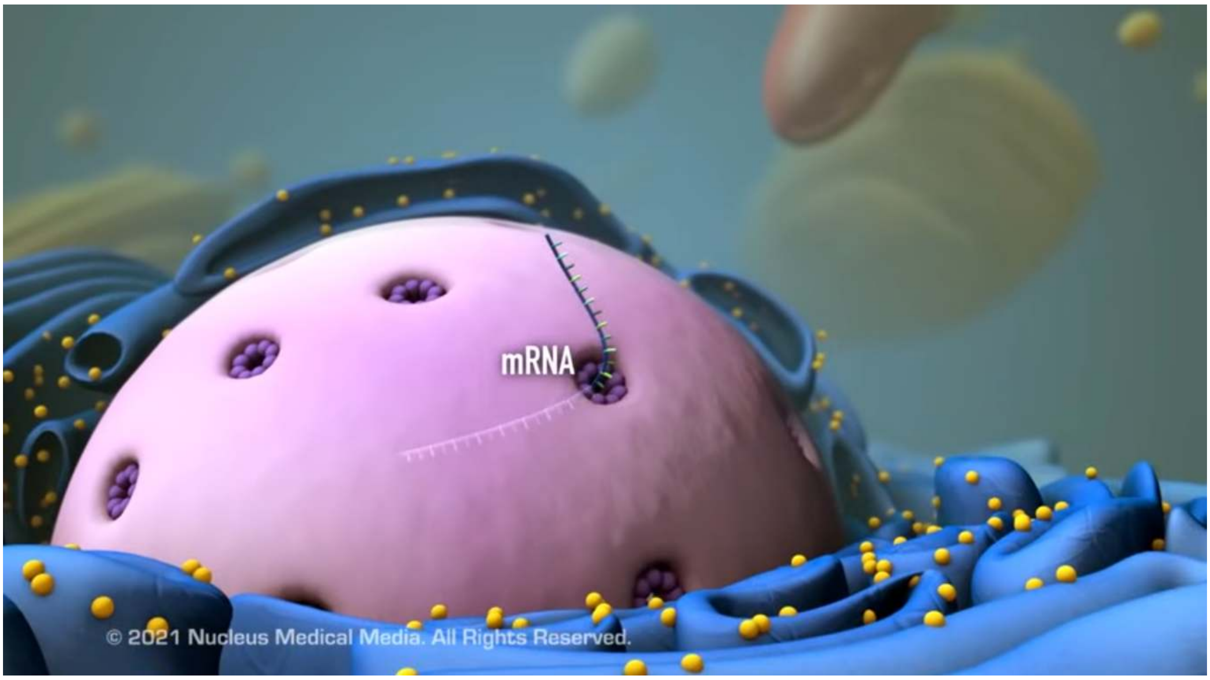
- on mrna molecule
- Same as dna coding (other) strand
- But has Uracil for thymine
    - U for T

- 

- Eukaryotes:
    - “Editing” or splicing of pre mrna molecule
    - Introns
        - Not code for proteins
    - Exons
        - Code for proteins
        - “Ex” → expressed
    - Introns spliced out
    - Exons stuck together

3. mRNA out of nucleus

- Mrna → out of nucleus → cytoplasm
    - Through nucleus pores

- 

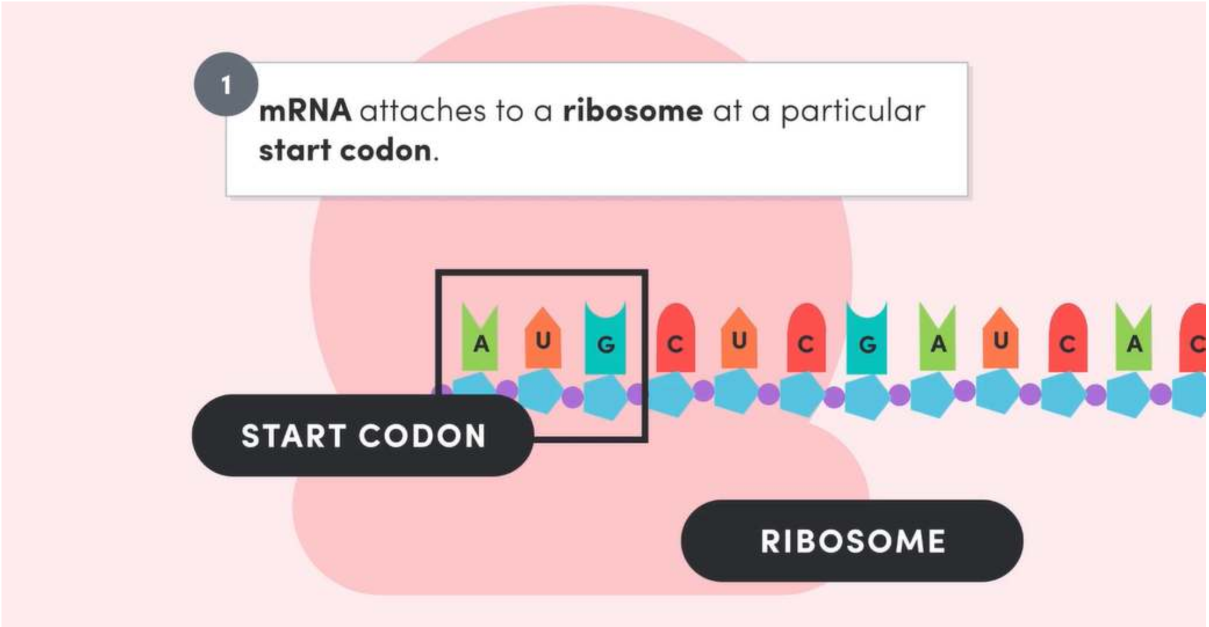
- Read by RIBOSOMES
- One mrna molecule
    - Read by large number of ribosomes
- Multiple chains of
    - same polypeptide product
        - Made from 1 mrna template molecules

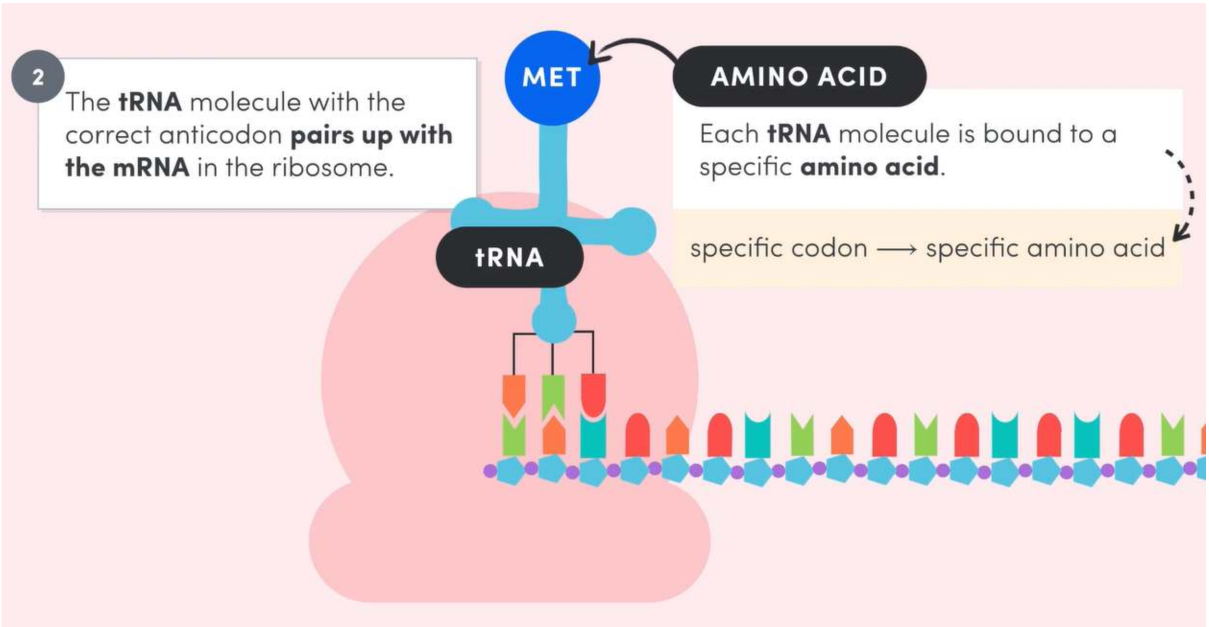
Translation:

- Turning mrna code → proteins by
    - assembling + linking amino acids

1.

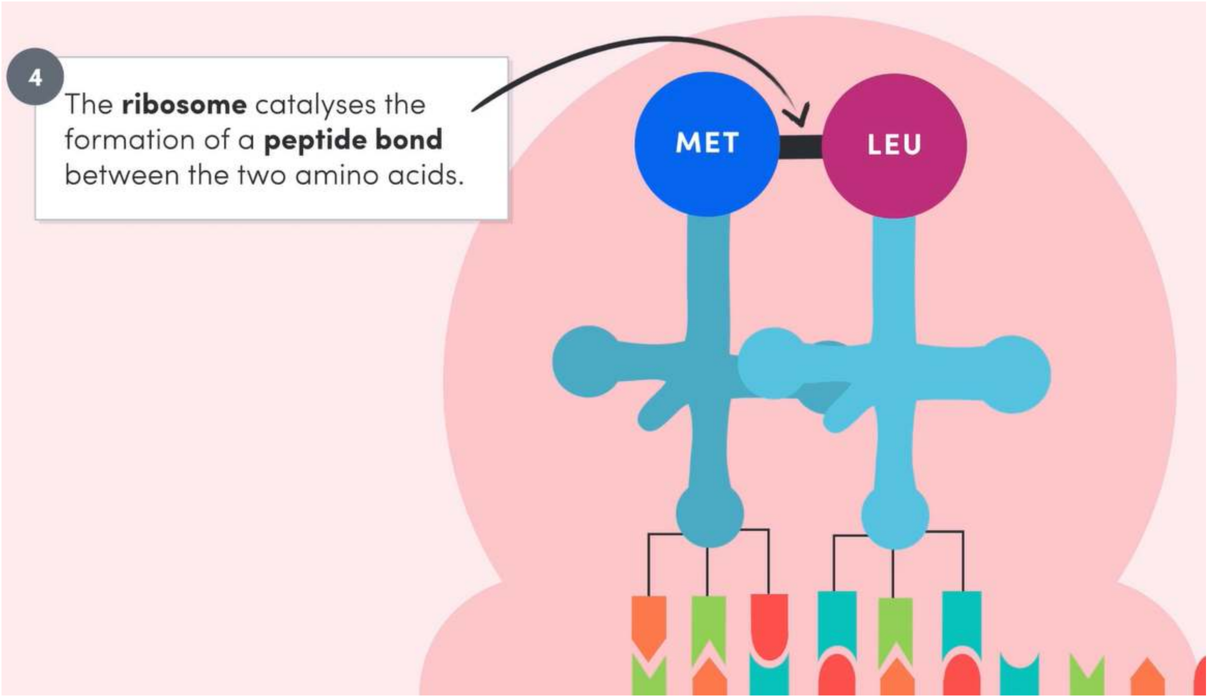
- Ribosomes move along mrna molecule
- Attach tRNA (transfer rna) molecules
    - to mrna by
        - Temporarily pairing bases of trna anticodons
            - To start codons on mrna
- Each trna SPECIFIC TO AN AMINO ACID
- Codons:
    - Set of 3 bases that
        - Make 1 specific amino acid
            - In polypeptide chain
    - Stop codons
    - Start codons

- 

- 

2.

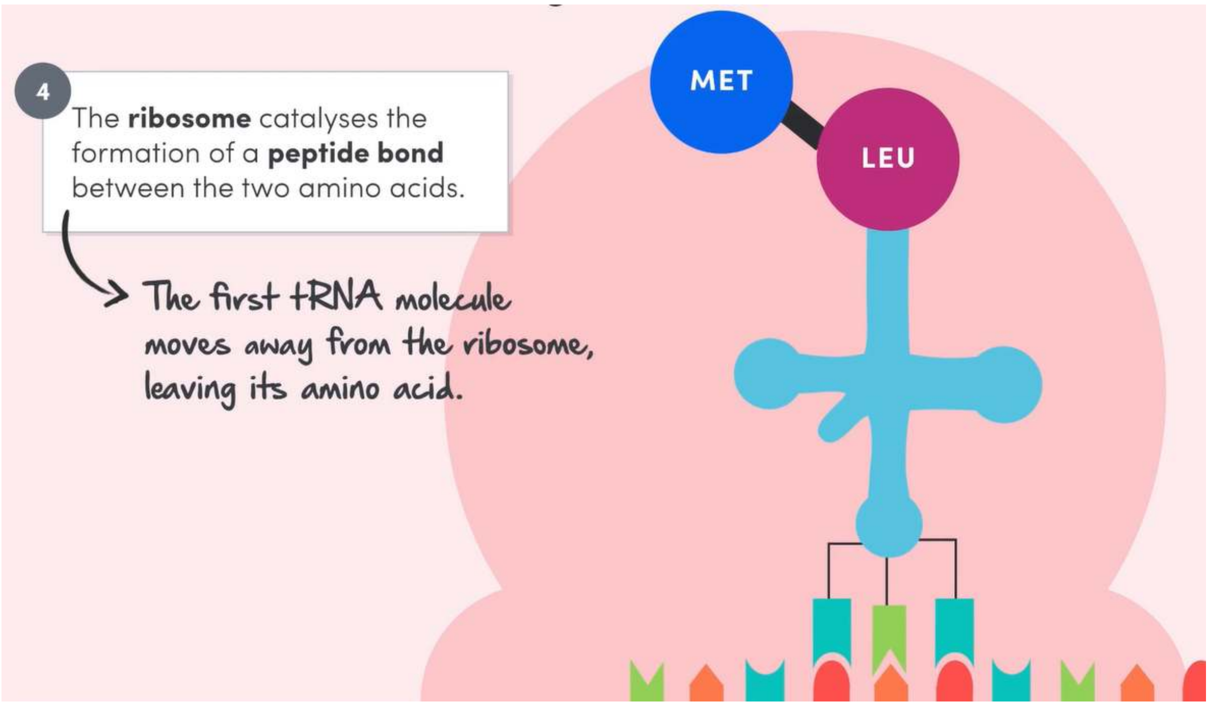
- Amino acids from tail of each trna
    - Linked to each other
        - By enzyme
            - to make polypeptide chain

- 

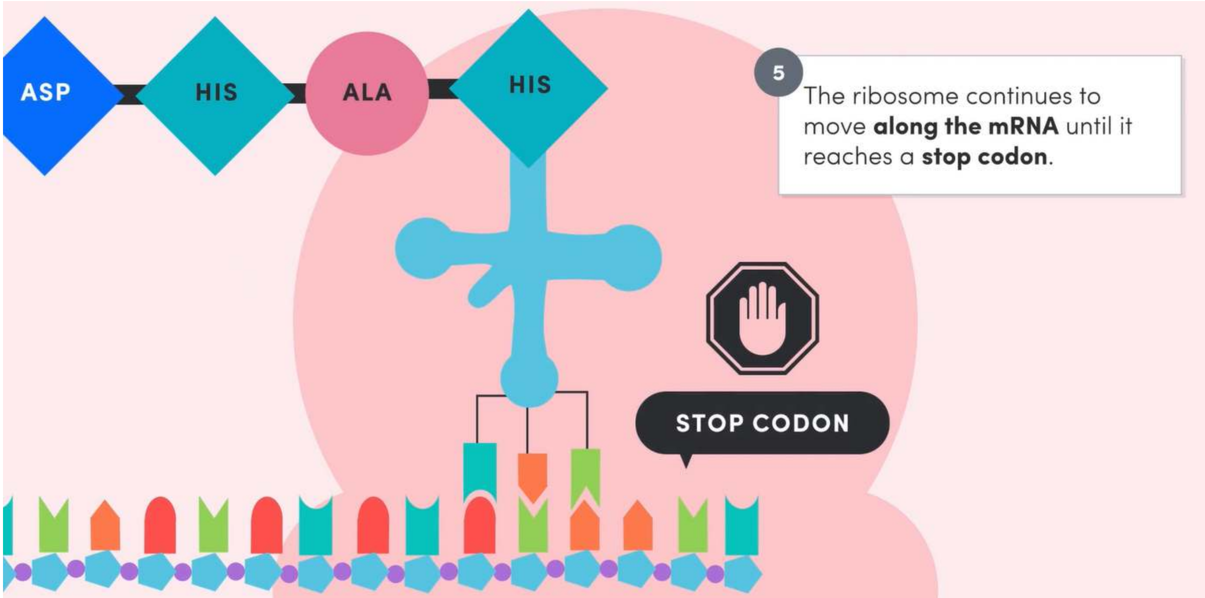
- Each amino acid
  - Spliced off its trna carrier

3.

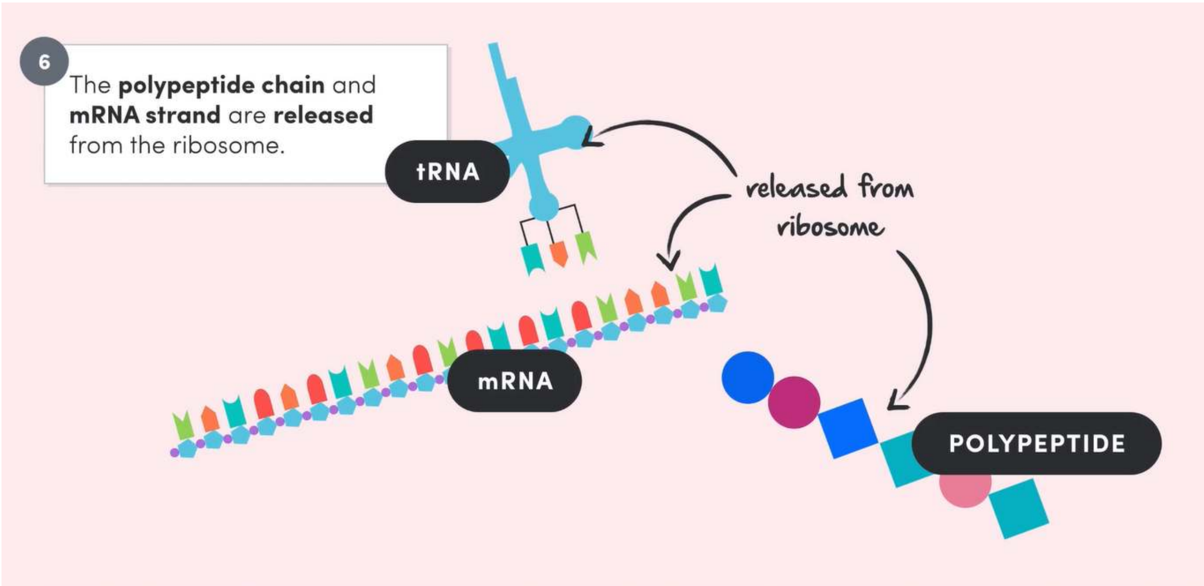
- Trnas move away from
  - Mrna
- Leave growing chain of amino acids
- Move back to cytoplasm
  - To pick up more amino acids
    - To be reused

- 

4.

- Ribsome continues moving along mrna
- Adding amino acids on mrna
    - Using trna
- Until reaching stop codon
- 

5.

- Polypeptide chain released from ribosome
- Trna attached to ribosome also released
- 

6.

- Polypeptide chain
    - processed in cell
- May be joined to 1 or more
    - Other polypeptides
        - Being processed further

- + folded into correct shape
  - For
    - Functioning
    - + forming final protein end product

7.

- Mrna breaks down into individual nucleotides
  - To be reused

RNA Processing:

- Pre mrna
  - Initial transcript
    - Having introns + exons
  - Introns:
    - Non-coding regions
    - Removed
  - Exons:
    - Coding regions
- Splicing
  - Process where
    - introns
      - Are removed
    - Exons
      - Joined
      - To make mature mrna
- Introns
  - Have signals
  - For removal
    - during splicing
- Alternative splicing:
  - Allows for different combinations
    - Of exons
      - To be joined
  - Makes
    - Different nature mrna molecules
      - From same gene
  - Leads to
    - Making of different proteins

- Same gene sequence
- In humans
  - Allows for
    - Rapid production
      - Of antibodies
      - In response to infections
- Are structurally or functionally similar
  - Called iso proteins

Importance of mRNA and tRNA in Transcription & Translation:

mRNA:

- Carries information
  - From dna in nucleus
    - → ribosomes in cytoplasm

tRNA:

- Has anticodon
  - That pairs with rna codons
- Binds to specific amino acid
  - Ensures correct sequence
    - during protein synthesis

Function and importance of Polypeptide Synthesis:

- Allows for making of proteins
  - For cell function
    - ∴ organism function

How Genes and Environment affect Phenotypic Expression:

- Phenotype refers to
  - An organism's observable characteristics
    - Not only physical appearance
- Genotype:
  - Genetic blueprint
  - Identical in all cells
- gene expression:
  - Controls cell specialisation

- Leads to
    - Production of proteins
        - Determine features of
            - Cells
            - + organism
- Variation in traits:
    - Due to
        - Genetics
            - nature
        - Environment
            - Nurture
        - Interaction between both
- Environmental factors
    - Can influence how genes are expressed
        - Not just genes
    - May cause chemical modifications to dna
        - w/o changing dna sequence
            - Affects gene expression
        - Modifications called
            - Chemical markers or tags
                - Can influence phenotype
- Identical twins
    - Often show phenotypic differences
        - Due to environmental influences
- Epigenetics
    - Field of studying how the environment
        - Can effect gene expression
            - At molecular level

Environmental effects on Gene Expression and Phenotype:

- Example: Siamese cats:
    - Show dark pigmentation at
        - Cooler parts of body
            - Due to temperature sensitive allele ©
    - Pigment only forms at low temps
        - Extremities like
            - Ears tail

        - Warmer body areas
            - Lack pigment
                - Due to inhibited gene expression
        - Phenotypic colour expression
            - Temperature dependent
- Hydrangeas:
    - Different flower colours depending on soil pH
    - Acidic soil
        - Blue flowers
    - Alkaline soil
        - Pink flowers
- Temperature + pH:
    - Alter shape of enzyme active sites
        - Influencing gene related
            - biochemical reactions
- Human height + weight:
    - Show interaction of genes with environment
    - Genetics influence traits
    - nutrition + toxin exposure
        - Modify outcomes
- Phenylalanine (PKU):
    - Genetic disorder
        - Where diet (environment)
            - Impacts gene expression
    - Phenylalanine buildup
        - Causes developmental issues
        - Low phenylalanine diet
            - Prevent severe symptoms
                - By influencing gene expression

Structure and Function of Proteins in Living Things:

Role of Proteins:

- Most abundant organic molecules in cells
- Make
    - Cell structures
    - Control chemical reactions

Structure of Proteins:

- Proteins made of
    - Long chains of amino acids
        - (polypeptides)
- Each protein
    - Folds into specific shape
        - Critical to its function
- Shape and chemical properties
    - Determine functions
    - Properties:
        - Charge
        - Water affinity

Chemical Structure:

- Contain:
    - Carbon
    - Hydrogen
    - Oxygen
    - Nitrogen
    - Sometimes sulfur
- Around 20 amino acids form proteins
- Amino acids
    - Joined by peptide bonds
- 1 or more polypeptides
    - Twisted together into particular shape
        - Results in overall structure
            - Of protein
- Sequence + arrangement
    - of amino acids
        - Determines proteins function

Structure of Proteins:

- Primary structure:
    - Linear sequence
        - Of amino acids
- Secondary structure:
    - Localised folding

- Local 3d shapes
  - Alpha helix
  - Beta sheet
- Result of + held by
  - hydrogen bonds

- Tertiary Structure:
  - More complex 3d folding
    - From disulfide + other bonds
  - E.g.:
    - Immunoglobulin
  - Caused by interactions between
    - Polypeptide + immediate environment
- Quaternary structure:
  - 2 or more polypeptides joined
    - To form larger functional protein
  - E.g.:
    - Haemoglobin
      - 4 separate peptides

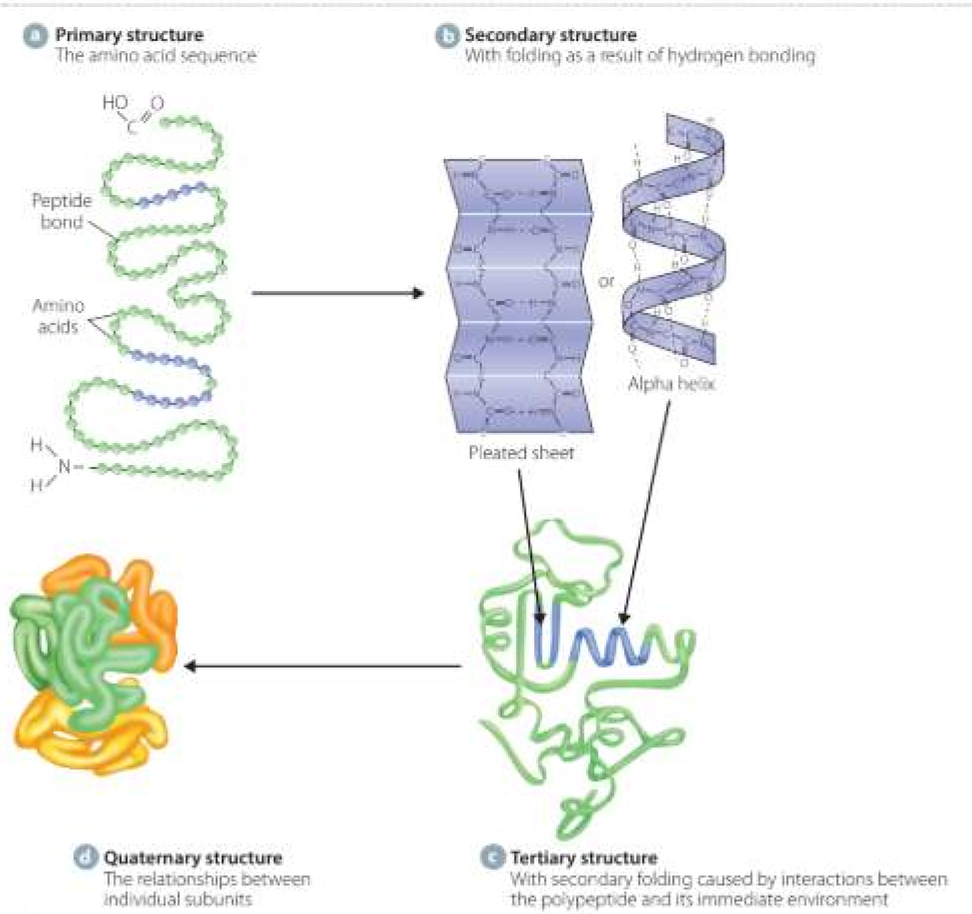


FIGURE 4.24 The four levels of protein structure: **a** primary, **b** secondary, **c** tertiary and **d** quaternary

-

Types of Proteins in Cells:

Fibrous:

- Provide structural support
    - In cells + tissues
- Combine with water
    - To form cytoplasm (cytoplasm)
- Long
- Stringy
- Insoluble in water
- E.g.:
    - Collagen
        - Strong
        - Coiled
        - In skin
    - Elastin
        - Skin

    - Ligaments
    - Tendons
  - Keratin
    - Hair
    - Nails

Globular Proteins:

- Spherical
- Compact
- Soluble in water
- Function as
  - Transport
  - Hormones
  - Enzymes
- E.g.:
  - Haemoglobin
    - Transports oxygen → blood
  - Immunoglobins
    - Antibodies
  - Various enzymes, hormones

Genetic Variation:

Meiosis, Fertilisation and Mutations:

Meiosis:

- Maintains constant
  - Chromosome number
    - From one generation to next
- Parent cell
  - Splits into 4 daughter cells
    - Instead of 2 daughter cells
      - Halving total chromosome number

Creation of Genetic Variation:

- Crossing over:
  - Chromosomes

- Line up in homologous pairs
    - 1 maternal
    - 1 paternal chromosome
- Exchange genes with each other
    - By wrapping arms around each other
        - Arms of chromosomes break
            - → exchange chromosomes
    - Mixes maternal and paternal genes
        - Creates genetic variation
- Ensures not all matching genes
    - On a chromosome
        - Are inherited together

- Independent assortment:
    - Moving of one chromosome
        - from each chromosome pair
            - → daughter cell
    - Halves the chromosome number in gametes
    - Creates genetic variation:
        - Depends on which chromosome
            - Maternal or paternal
                - Of each pair
                    - Goes into which daughter cell
    - ∴ makes different combinations of genes
        - In different gametes
- Each resulting daughter cell
    - Is different
        - As combination of
            - Maternal + paternal
                - Chromatin is different
- Variety of gametes
    - All with different makeup
        - Form
- Fertilisation:
    - Different combinations of gametes
        - May fuse
            - Having potential to create
                - Completely genetically different organisms

- Variation may also result:
    - From mutation
        - At any point in meiosis
            - Most commonly in
                - Dna replication

Fertilisation and New Combinations of Genotypes:

- Many combinations
    - Of gametes
        - that may fuse
    - Having potential to create
        - Completely genetically different organisms
- Depends on which gametes
    - fuse with each other

Self and Cross Fertilisation:

- Cross fertilisation:
    - Different individuals of same species
    - Greater genetic diversity
- Self fertilisation:
    - 1 individual of a species
        - Having male and female organs
    - Less genetic diversity
- Unisexual species:
    - Individuals having organs of
        - 1 sex
        - Opposite sex individuals
            - Fertilise gametes
    - Greater genetic diversity
- Bisexual species:
    - Individuals having organs of
        - both sexes
        - One individual
            - Uses both sets of gametes from organs
    - Less genetic diversity

Genotypes and Inheritance Patterns:

- Gregor Mendel

Autosomal Recessive Inheritance:

- Happens under conditions:
    - A version of each characteristic or trait in individual
        - Inherited from both parents
        - Controlled by a pair of inherited factors
            - "Alleles"
    - Alleles pass from
        - 1 gen → next gen
            - According to set ratios
    - Alleles in individual may be
        - Same
            - Homozygous
            - "Pure bred"
        - Different
            - Heterozygous
    - In hybrid people
        - Trait expressed
            - Is dominant allele
        - Trait hidden
            - Is recessive allele
    - For recessive trait expressed
        - Both alleles in person
            - need to be recessive
    - When gametes form
        - Pair of alleles for a trait
            - Separate
    - Each gamete gets
        - One allele for trait/gene
    - If inheritance of one or more traits/genes studied
        - Pairs of alleles for each trait
            - Separate independently of other pairs of alleles
    - Assumes alleles are
        - Located on non sex chromosomes

- autosomes

Modern Genetics Technology used to Describe Inheritance Patterns:

- Different genes
  - Control different traits
- Genes
  - Segments of dna on chromosomes
    - determine inherited characteristics
    - By providing code for producing specific proteins to be expressed
      - E.g.:
        - Eye colour
        - Hair colour
- Alleles:
  - Different forms of same gene
  - Occur in pairs
    - In diploid organisms
  - Located at same locus (position)
    - On homologous chromosome pairs
  - In some populations:
    - May be more than 2 alleles for a gene
    - Individuals can only have
      - 2 alleles
  - Alleles possessed
    - Depend on pair of alleles
    - Passed down from parents
- Diploid cells
  - Have 2 alleles for 1 gene
- Haploid cells
  - Have 1 allele for 1 gene
- Phenotype
  - Observable traits
  - E.g. appearance of an organism
  - Expressed gene of organism
  - May include
    - Physiology (functioning)
      - Of organism
  - Sum of gene products (proteins + rna) that are made

- Genotype
    - Genetic makeup
        - Or combination of alleles in organism

-

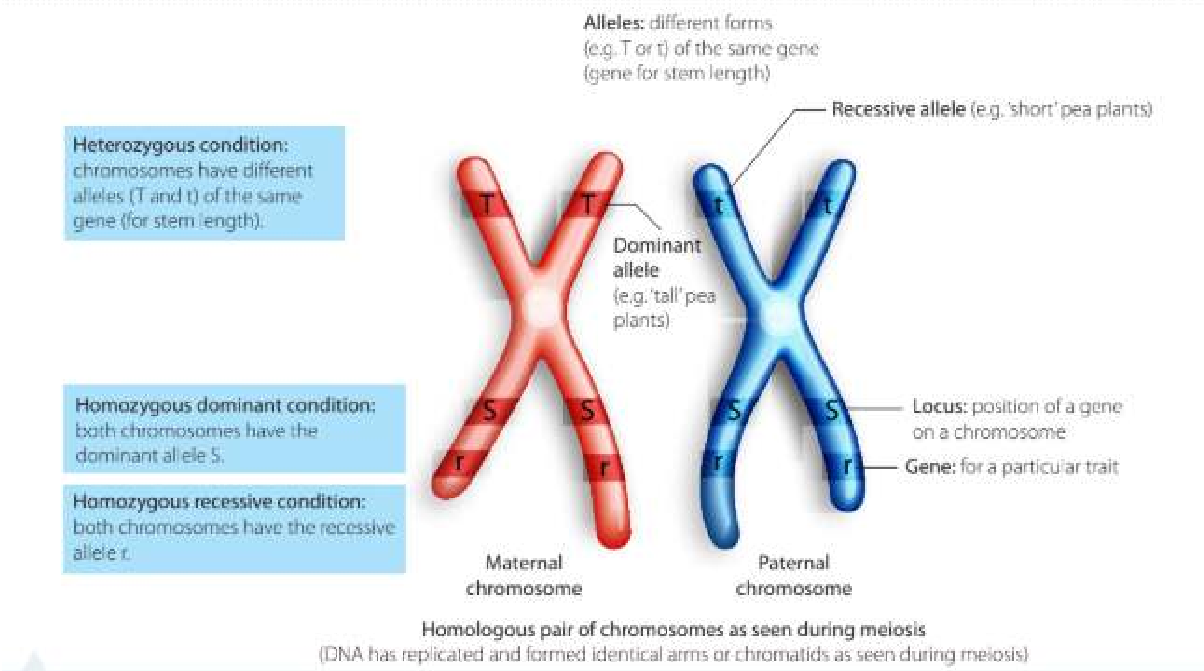


FIGURE 5.8 Pairs of chromosomes showing alleles, gene, locus, and homozygous and heterozygous genotypes

Autosomal Recessive Inheritance and Genetic Crosses:

How Mendel's pea plants helped us understand genetics - Hortensia Jiménez Díaz

- Mendel bred
    - Yellow purebred plant + green purebred plant
        - Purebred = homozygous genotype
            - Homozygous
                - Dominant
                - Or recessive
        - Yellow = dominant
        - Green = recessive
    - If pure Y + y breed to make offspring
        - Offspring = Y + y = Yy
            - (heterozygous)
        - All offspring
            - Were yellow as
                - They had a dominant Y allele
                    - Which was expressed
- Then bred

    - Heterozygous offspring together
        - Green and yellow
            - Expressed
- 1st gen punnet square:
    -

	Y	Y
y	Yy	Yy
y	Yy	Yy

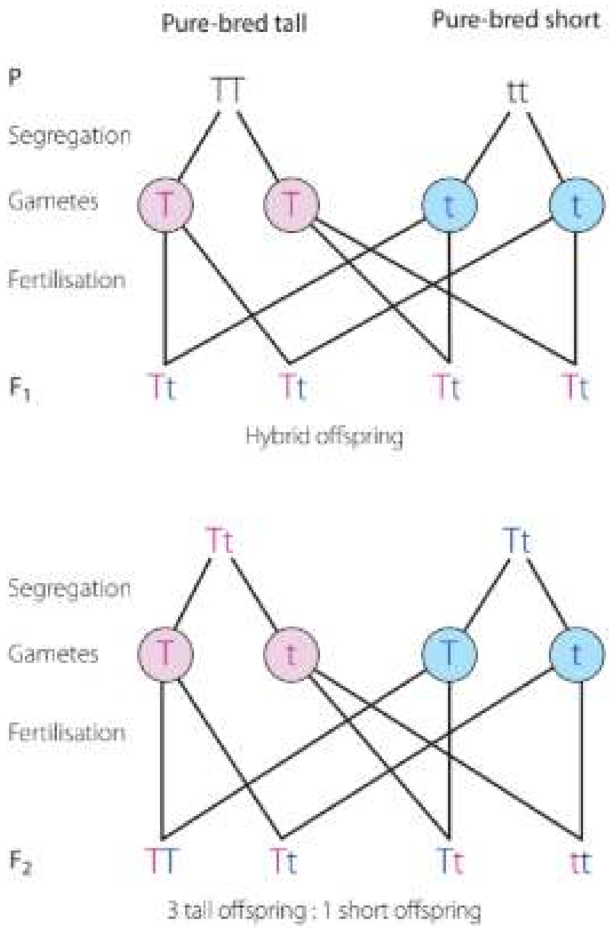
    - All offspring Yellow as they have
        - 1 dominant Y allele
- 2nd gen punnet square:
    -

	Y	y
Y	YY	Yy
y	Yy	yy

    - Genotype ratio:
        - 2Yy : 1YY : 1yy
    - Phenotype ratio:
        - 3 Yellow : 1 Green
    - As heterozygous genotypes bred
        - Alleles become mixed
        - Resulting in
            - 3 yellow offspring
                - With dominant Y allele
            - 1 green offspring
                - Only expressed with 2 recessive y alleles

Mendels Monohybrid Cross:

- For short and tall pea plants



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Mendel's Laws:

Law of Segregation:

- Genes and alleles
  - Separate randomly during
    - Making of gametes
- Gametes have
  - 1 allele represented only
- Offspring
  - Inherit 1 allele
    - From each parent

Law of Dominance:

- In purebreeding cross
  - For contrasting traits
    - TT * tt → Tall with short
  - Only 1 form of trait will appear
    - In next gen

- Hybrid offspring:
    - Have only 1 dominant trait
        - In phenotype
- If 2 hybrids breed
    - Ratio of 3 : 1
        - Dominant to recessive traits
        - (seen in yellow green pea punnet square)

Law of Independent Assortment:

- Dihybrid crosses
    - One or more
        - Crossing of traits
- Genes for different traits
    - Inherited independently
        - of each other
- Means that
    - Traits are not related
        - Or their inheritance
            - Not influenced by each other
- ∴ Assumes that
    - Genes are located on
        - Different chromosomes

Solving Genetics Problems:

Punnet Square:

-

	Y	Y
y	Yy	Yy
y	Yy	Yy

- Square cross showing
    - All possible genotype combinations
        - During fertilisation
- Shows how Alleles are segregated
    - In gametes
        - During meiosis

Probability:

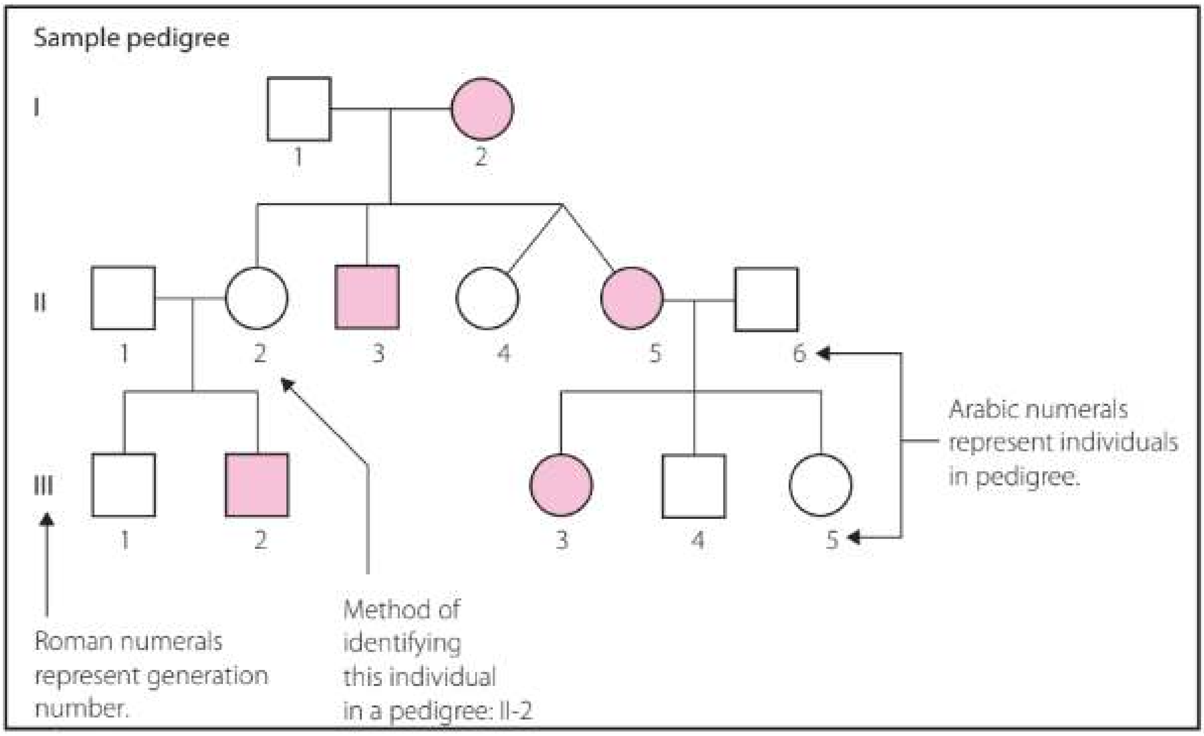
- Percentages of traits
    - From punnet squares

Test Cross:

- Used to determine
    - Whether an organism
        - Showing a dominant phenotype is
            - Homo (VV)
            - Or heterozygous (Vv)
- Involves crossing
    - unknown genotype
        - With homozygous recessive (vv) organism
- Example:
    - Drosophila wing length
    - V = dominant allele for long wings
    - v = recessive allele for short (vestigial) wings
    - Cross 1:
        - VV * vv
        - All offspring are Vv (heterozygous)
        - 100% have long wings
        - ∴ tested parent
            - homozygous dominant
    - Cross 2:
        - Vv * vv
        - Offspring are
            - 50% vv (short)
            - 50% Vv (long)
        - 1 : 1 phenotypic ratio
        - ∴ tested parent
            - heterozygous dominant

Pedigree Analysis:

- Square = male
- Circle = female
- Shaded individuals
    - Have trait being tracked

- Roman numerals
    - For each generation
- Linking lines:
    - Marriage
        - horizontal line
    - Offspring
        - Vertical line
- Pedigrees used to:
    - Determine inheritance patterns
    - Assign genotypes
        - to individuals
            - If possible
    - Make predictions about probability (or risk)
        - Of individual inheriting a trait
            - E.g.:
                - Genetic disorder
                - Abnormality
                - Disease
- 

- If referring to specific individual:
    - Use format
        - "Gen no." - "individual number"

Autosomal Dominant Inheritance:

- Basically all generations
    - affected
- Inherited from mom or dad
- Males + females
    - Affected at same rate
- An affected individual
    - ALWAYS has 1 affected parent
- 2 unaffected parents
    - May have unaffected child
- Example disorders:
    - Huntington's disease
    - Polycystic kidney disease

Autosomal Recessive Inheritance:

- Inherited from
    - 2 unaffected parents
- 2 affected parents
    - Only have affected children
- Not every generation affected
- Both recessive alleles
    - Need to be present for disorder
- Example disorders:
    - Cystic fibrosis
    - Sickle cell anemia

X-Linked Recessive Inheritance:

- Can be inherited from
    - 2 unaffected parents
- If father affected
    - Daughter can be affected
    - If mom is also affected
        - As 1 X (guaranteed affected from dad)
        - And 1 X (maybe affected from mom)
- Male affected more
- If mother affected
    - (has both recessive alleles)

    - Son is guaranteed affected
- All sons of affected mother
    - Are affected

Solving Pedigrees:

- Autosomal dominant:
    - Affected children
        - have affected parents
- X linked recessive:
    - Mom affected
        - ALL sons MUST be affected
- Autosomal recessive:
    - Unaffected parents
        - Have affected children
    - BOTH affected parents
        - MUST have ALL affected children
    - example

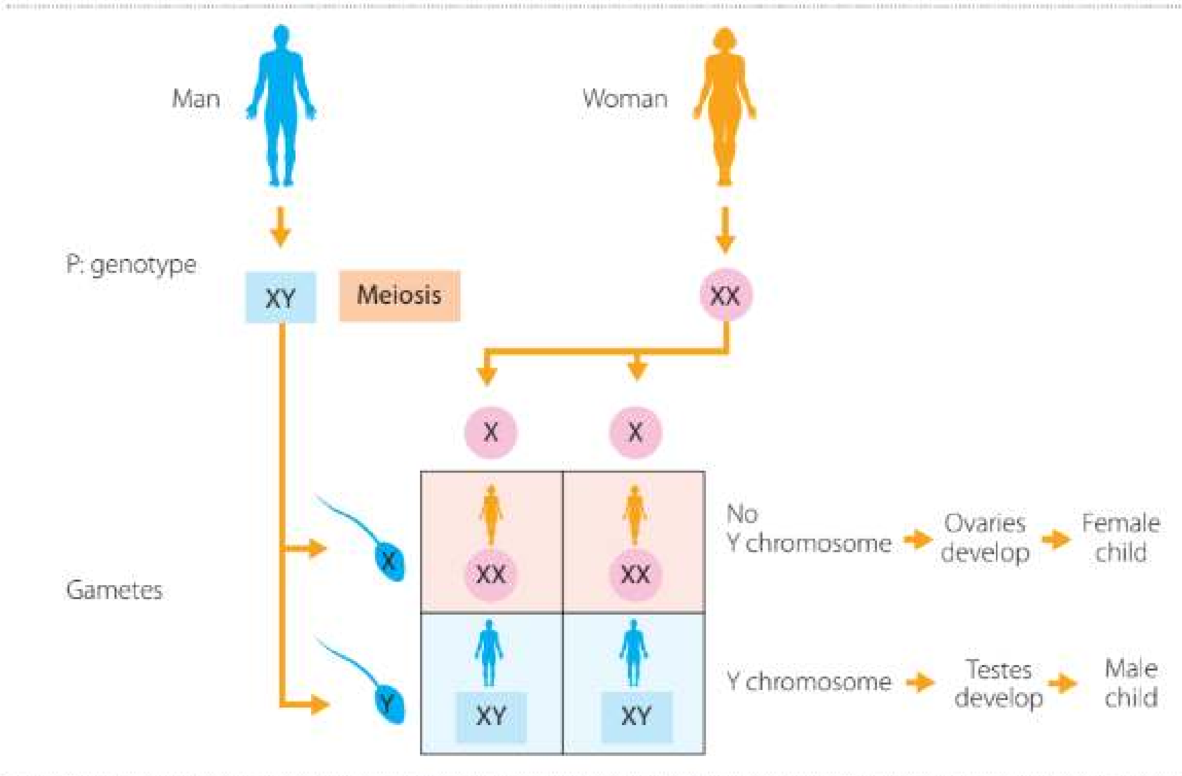
Deviations from Mendel's Ratios:

- Sex linked inheritance
- Codominance
    - 2 traits expressed
        - Not dominant or recessive
- Incomplete dominance
    - Not 1 trait fully expressed
    - Blending of traits expression

Sex Determination:

- Sex chromosomes
    - Separate during
        - meiosis
    - Recombine during
        - Fertilisation
- Determine whether offspring
    - Male
    - Female
- During meiosis

- Sex chromosomes segregate
- Females:
  - Diploid
    - 44 autosomes + XX
  - Haploid:
    - 22 autosomes + X
    - In all eggs
- Males:
  - Diploid:
    - 44 autosomes + XY
  - Haploid:
    - 50% of sperm
      - 22 autosomes + X
    - Other 50% of sperm
      - 22 autosomes + Y
- Fertilisation
  - Recombination of sex chromosomes
    - From gametes
- X from mom, X from dad
  - XX
  - Female
- X from mom, Y from dad
  - XY
  - Male



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Sex Linkage:

Recessive:

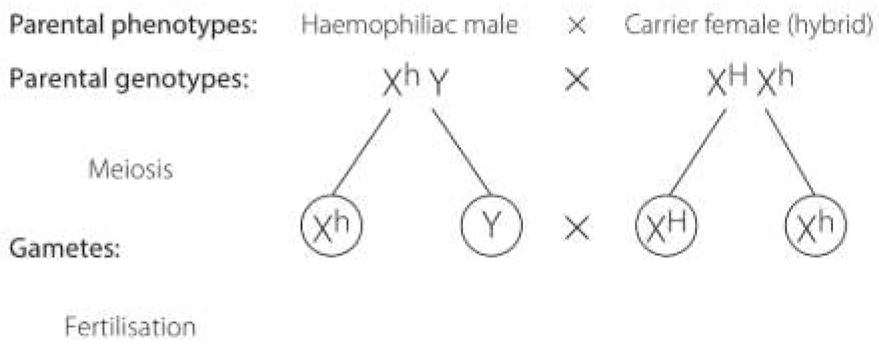
- Some genes carried
    - on X, Y chromosomes
        - Code for characteristics
            - Other than gender of individual
- If disorder gene on X chromosome
    - Females have 2 alleles
        - Due to XX
            - Recessive affected allele X
            - Can be masked by
                - dominant unaffected allele X
                - ∴ unaffected but carrier
    - Male has 1 allele
        - Due to 1 X
            - ∴ if X affected
                - = guaranteed disorder
- Recessive disorders
    - More seen in males
    - More likely in X-linked disorders

- Affected females
    - Pass disorder → sons
- Affected males
    - May pass disorder → daughters
        - As one X comes from them → daughter
        - But dependent on mother affected
- Example X linked recessive disorder:
    - Haemophilia

Symbols Used:

$$X^H X^H = \text{normal female}$$
$$X^H X^h = \text{carrier female (heterozygous)}$$
$$X^h X^h = \text{haemophiliac female (homozygous = lethal)}$$
$$X^H Y = \text{normal male}$$
$$X^h Y = \text{haemophiliac male}$$

-



Punnett square

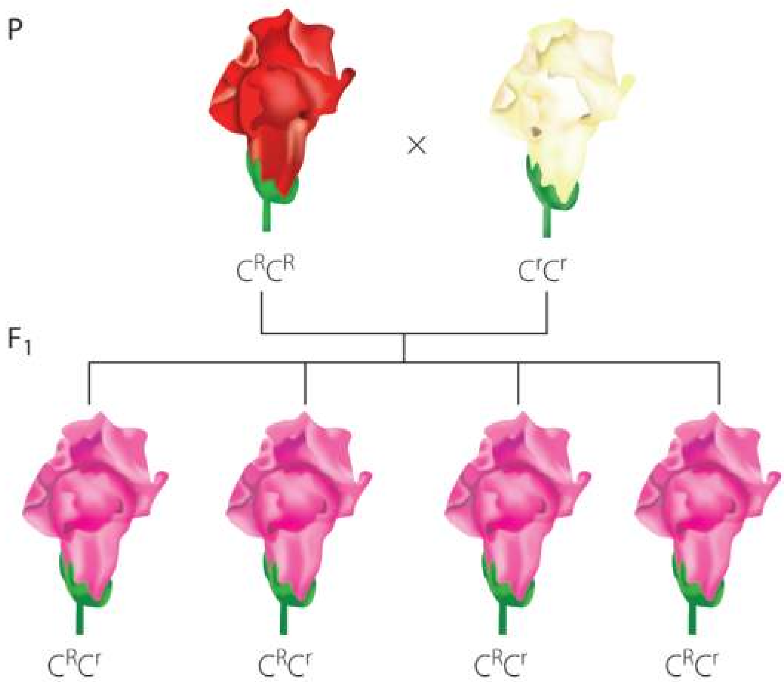
Gametes	X^H	X^h
X^h	$X^H X^h$	$X^h X^h$
Y	$X^H Y$	$X^h Y$

-

Incomplete Dominance and Codominance:

Incomplete Dominance:

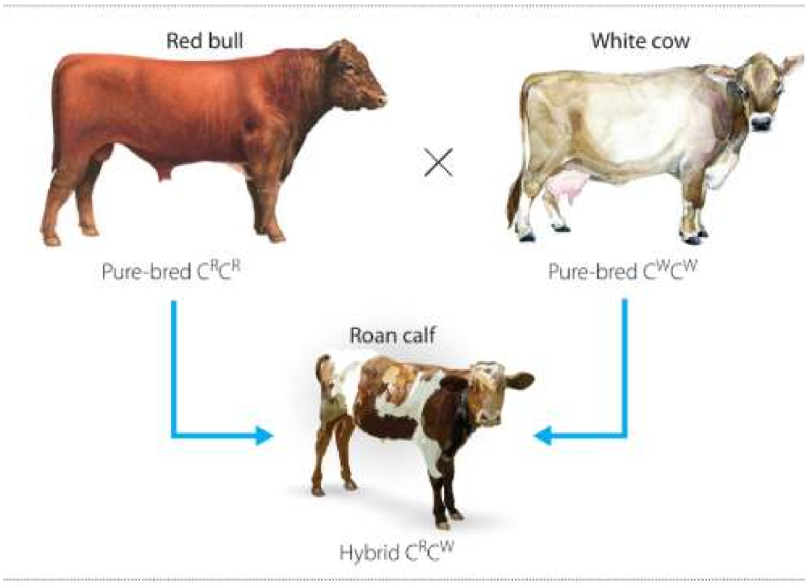
- Blending of features
    - Of 2 alleles expressed

- Gives
    - Hybrid that is intermediate
- Not one allele expressed fully
- "In between" phenotype
- Example:
    - Red snap dragon flowers
        - Crossed with
            - White flowers
                - Make pink flowers
- Symbols:
    - C for colour (trait)
    - Superscript for allele
    - C^R = red colour
- 

Codominance:

- Both alleles expressed
- Creates new phenotype
- Co = both
- Example:
    - Pure breeding cattle
        - Have red or white coat colour
    - Hybrid individuals
        - Have 1 allele for red

- + 1 allele for white
- Show both coat colours
  - Interspersed

- 

Multiple Alleles:

Blood Type:

- Blood types:
  - A
    - dominant
  - B
    - dominant
  - AB
    - codominant
  - O
    - recessive
- Blood type
  - Distinguished by different antigens
    - A antigens
    - B antigens
    - A + B antigens
    - No A + or B antigens
  - Is genetically inherited
- Notation:
  - Gene = I

- Allele = ^A, ^B

- Genotypes:
  - I
    - immunoglobulin
  - A:
    - I^A, I^A
    - I^A, I^i = i
  - B:
    - I^B, I^B
    - I^B, i
  - AB:
    - I^A, I^B
  - O:
    - i, i
      - As it is recessive

Polygenic Traits:

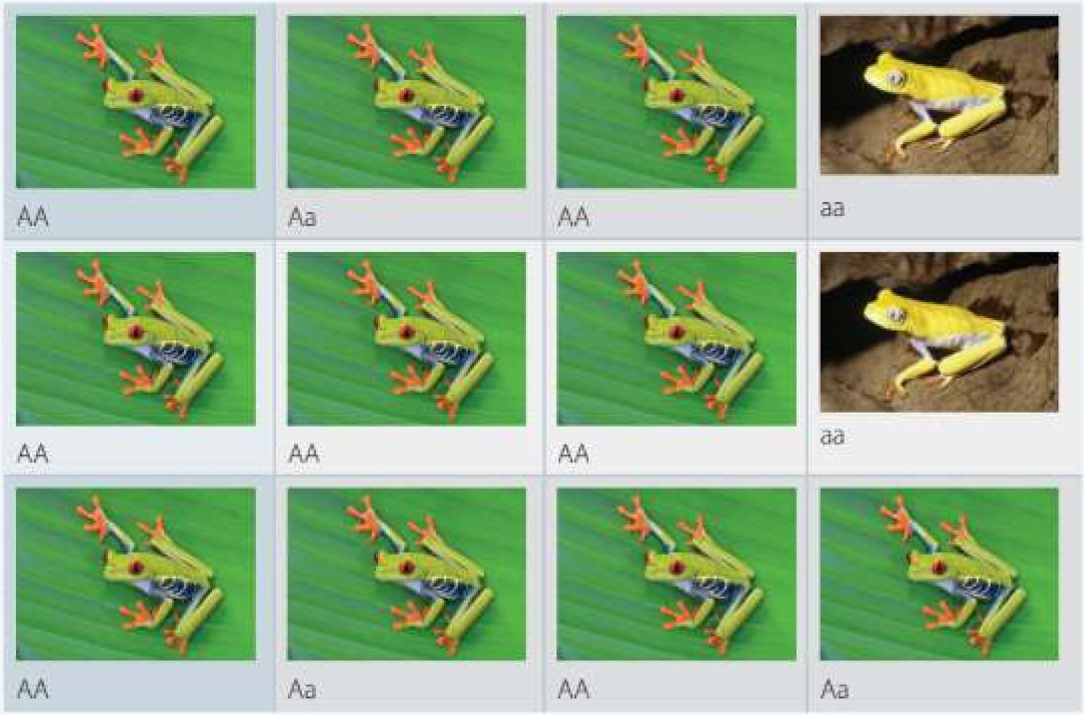
- More than 1 gene used
  - To express a trait
- Half genes from mom
- Half genes from dad
- E.g.:
  - Skin colour
  - Height
    - Both may be affected
      - by environmental factors
        - Genes stay same

Population Genetics:

- The study of
  - How the gene pool
    - Of a population
      - Changes over time
- Changes → species evolving
- Gene pool:
  - Total sum of all genes
    - + their alleles in a population

- Allele frequency
    - Measure of
        - How common
            - An allele is within a population
- Frequency of allele G
    - $$= \frac{Number\ of\ copies\ of\ allele\ (G)\ in\ population}{Total\ number\ of\ copies\ of\ alleles\ (G+g)\ in\ the\ population}$$
    - If multiple alleles (more than 2)
        - Total copies of gene
            - Add all different alleles together
    - E.g.:

TABLE 5.5a Frog population in first generation (generation 1)



        -
        - Allele frequencies:
            - A:
                - = Total A alleles in gen / Total of all all alleles in gen
                - = 17 / (12*2) [12 frogs *2 alleles for each frog] = 17/24
            - a:
                - = 7 / (12*2) = 7/24

Hardy-Weinburg Equilibrium:

- Calcs distribution of
    - genotype frequency

        - On non evolving population
- Theorem states that:

1.

- Allele frequencies in population
    - Not change
        - From gen → gen

2.

- If allele frequencies
    - in population with 2 alleles
- Genotype frequencies:
    - p^2
    - 2pq
    - q^2
-

	A XX gamete (p)	a XX gamete (q)
A XY gamete (p)	AA (p^2)	Aa (pq)
a XY gamete (q)	Aa (pq)	aa (q^2)

- ∴ freq of
    - AA = p^2
    - Aa = 2pq
    - aa = q^2

3.

- If 2 alleles
    - p + q = 1

Assumptions:

1.

- Natural selection
    - Does not act on population

- Means:
    - No consistent differences
        - In probabilities of survival or reproduction
            - Among genotypes

2.

- Mutation
    - (new alleles)
- + migration
    - (movement of individuals + their genes in or out of population)
        - Don't add alleles into population

3.

- Population size
    - Infinite
- Means genetic drift
    - Not causing random changes
        - In allele frequencies
            - Due to
                - Sampling error
                    - From 1 gen → next

4.

- All mating
    - Random

Equations:

- Genotype freq.:
    - $p^2 + 2pq + q^2 = 1$
- Allele freq.:
    - $p + q = 1$
- Where
    - p = freq of dominant allele (A)
    - q = freq of recessive allele (a)
    - p^2 = freq of
        - homo dominant genotype
        - AA
    - $2pq$ = freq of
        - hetero genotype
        - Aa

- q^2 = freq of
    - homo recessive genotype
    - Aa
- E.g.:
    - Cystic fibrosis is a recessive condition that affects about 1 in 2,500 babies in the Caucasian population of the United States. Calculate the following.
    - A. freq of recessive allele
        - $q^2 = 1/2500$, $q = \sqrt{1/2500} = 1/50 = 0.02$ (2%)
    - B. freq of dominant allele in population
        - 1 - 0.02 = 0.98 = 98%
        - As cystic fibrosis recessive condition
            - To get, need aa (both recessive)
        - If only 2% have it
            - Everyone else has dominant allele as
                - Recessive allele is not expressed
    - C. % of heterozygous individuals in population
        - 2pq = heterozygous genotype
        - p = 0.98, q = 0.02
            - Use prev parts of question
        - 2(0.98)(0.02) = 0.392 = 4%

Single Nucleotide Polymorphisms (SNPs):

- Polymorphism
    - “Many forms”
    - Genetic variations that
        - cause different phenotypes
    - From mutations
        - Normally due to
            - Errors in dna replication
- Snp is
    - Variation at
        - Single nucleotide position
            - In dna
                - Of people
    - When 1 nucleotide
        - Replaced by another

- During dna replication
- Change must happen in
  - At least
    - 1% of population
      - To be classed as a snp
        - (Not only mutation)
- Snp happens
  - Specific locations on chromosomes
    - Can affect
      - Traits
      - Health
  - In non coding regions
    - Of dna

Why they are Important:

- Variations lead to
  - Differences in
    - Appearance
    - Disease susceptibility
    - Response to drugs
    - Enzyme function
- Most snps happen at
  - Non coding regions of dna
- Don't directly cause
  - Phenotypic changes
- Still valuable
  - Genetic markers

SNPs as Genetic Markers:

- Genetic marker
  - Is a known dna sequence
    - At specific location on a chromosome
      - Used to identify
        - Individuals
        - Traits
- Snps used to
  - Distinguish individuals

            - Genetically
        - Trace
            - inheritance patterns
        - Identify
            - Disease risk factors
- Human genome has
    - 10 million snps

Genome-Wide Association Studies:

- Gwas
    - scan genome
        - For snps
            - That happen more frequently in
                - People with a specific disease
                    - Than in those without it
    - Are powerful tools for
        - Identifying
            - Disease related genetic regions
        - Finding markers
            - For complex disease
        - Studying
            - Genetic risk factors
- Scientists analyse
    - Hundreds → thousands of snps
        - Making them
            - Efficient
            - + cost effective

            Over full genome sequencing

SNP Frequency and Grouping:

- Frequency:
    - About 1 in every 300 nucleotides
        - ~10 million SNPs
            - Per human genome
- Are often studied in groups
    - Haplotypes
        - Clusters of linked SNPs

            - Inherited together
        - Provide more powerful
            - Genetic Associations
                - Than individuals SNPs

Uses of Haplotypes and SNP Data:

- Indicators of disease
    - E.g.: diabetes, alzheimers
- Determining
    - family relationships
    - Ancestry
- Studying
    - Evolutionary relatedness
        - Among populations
- Building
    - Biobanks + databases for
        - Genetic research

Scientific and Medical Benefits:

- Enhances understanding of
    - How genetic variation
        - Contributes to
            - human health
            - + disease
- Supports development
    - Of personalised medicine
    - Where
        - Treatment is tailored
            - To individual's
                - genetic profile
- Allows researchers to
    - Detect low frequency polymorphisms
        - In populations
    - Access large scale
        - Genomic data
            - Through shared databases

Genotyping Vs. Sequencing:

- Genotyping:
    - Identifies specific genetic variations
        - Like SNPs
            - In individuals
    - Focuses on
        - Known differences
            - > whole DNA sequence
    - faster
    - + More cost effective
        - Then sequencing
- Sequencing:
    - Determines exact order
        - Of nucleotides (ATGC)
            - In dna strand
    - Detailed
    - Expensive
    - Time consuming

Inheritance Patterns in a Population:

DNA Technologies:

PCR and Gel Electrophoresis:

Polymerase Chain Reaction (PCR):

- Amplifies a section of dna
- Amplify:
    - Make many copies
        - Of dna section
- Used in vitro
- Vitro:
    - PCR being performed in test tube
        - > living organism
- Used for
    - Testing
    - Analysing

- Using

  DNA sample
- Only used for
  - Copies of dna

Materials:

- Dna sample
  - Copied for pcr
  - May be from
    - Crime scene
    - Person being tested
      - for genetic disease
- Free nucleotides
- Heat stable dna polymerase
  - Same enzyme for dna replication
  - But more heat tolerant
  - As pcr needs heat
    - And enzymes sensitive to temp + pH
- Primers
  - Act as signal
    - for dna polymerase
      - To copy that region of dna
- Buffer
  - Liquid where all other ingredients added
  - Prevents
    - Sudden pH changes

Process:

- Uses variations in temp
  - To control
    - Replication process

1. denaturation

- Pcr reaction mix
  - Heated to 95 degrees c
- Dna denatured
  - Separates to form 2 single strands

2. Annealing

- Pcr mix
    - Cooled to 55 degrees c
- Primers bind to dna

3. extension

- Pcr mix
    - Heated to 72 degrees c
- Dna polymerase
    - Moves down template
        - Synthesises new dna

4. Back to denaturation

- Pcr mix
    - Reheated to 95 degrees c
- Separates dna
- + stops dna synthesis
    - To repeat process

Gel Electrophoresis:

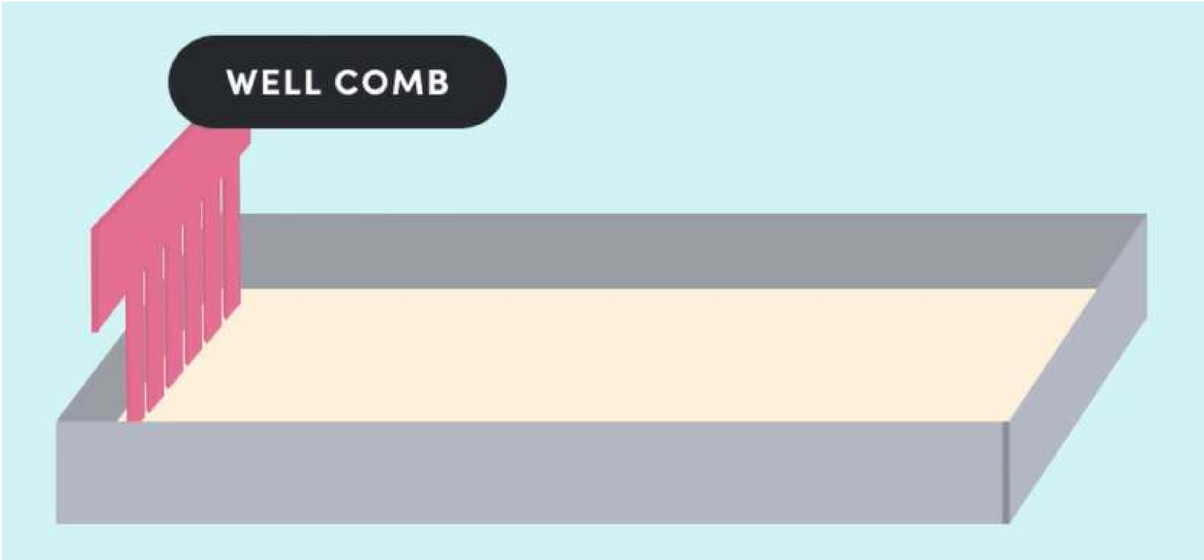
- Dna recap:
    - Dna has
        - 2 strands of nucleotides
            - Intertwine to form
                - Double helix
    - Sugar + bases are
        - Neutral
    - Phosphate
        - Negative
    - ∴ dna is negative

Process:

1. Prepare set up

- Pour liquid agarose gel
    - Into mould
- Put in well comb
- When mould set
- Put in gel box
- Gel box
    - Positive electrode 1 end

- Negative electrode other end

- Wells on
  - Negative side
- Immerse gel with buffer
  - Can conduct electricity
- 

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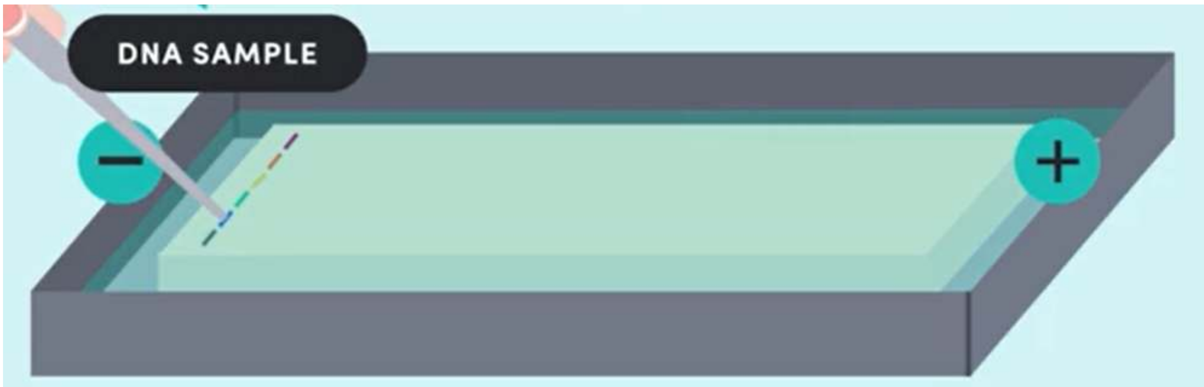
2. Put in dna samples + dna ladder

- Each different dna sample

- Put into its own well
- Samples amplified using pcr
  - Before being put in gel
- Dna ladder
  - Put into 1 well
  - Mix of known dna fragments
    - Each of which
      - Has specific size
- In end
  - Allows us to determine size
    - Of dna fragments
      - By comparing to ladder
- Record each dna sample in well
- 

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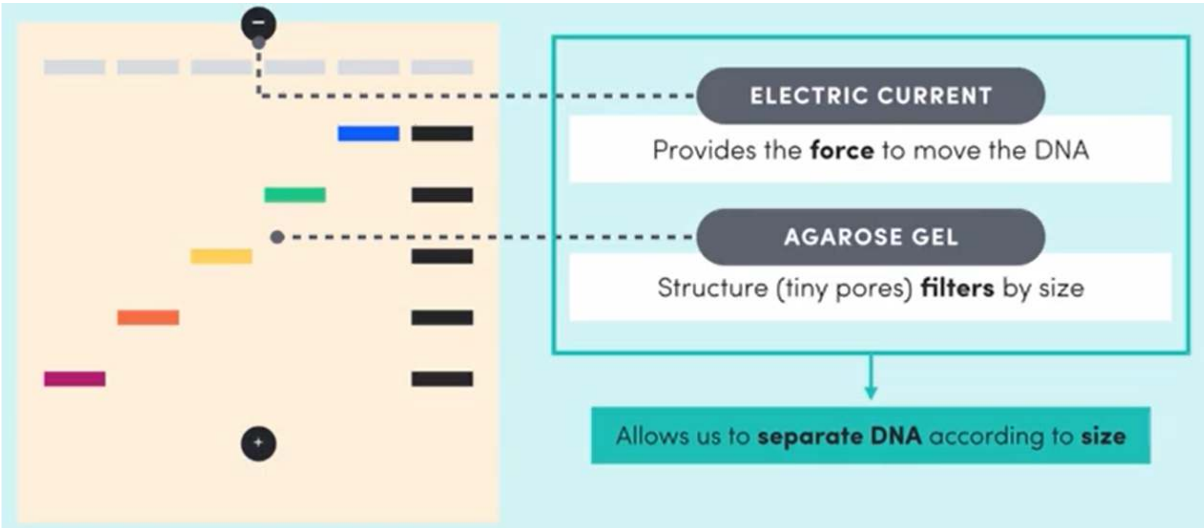
3. Run gel

- Turn power on
  - Current runs through gel
- Dna will move to positive side of gel
  - Due to it being negative
    - Opposites attract
- Dna filtered by size
  - Because of pores
    - In gel
  - Dna that is smaller

- Goes through pores more easily
    - ∴ moves faster
        - Towards pos side of box
- Electric current
    - Is force to move dna
- Gel
    - Small holes
    - Filter dna by size
- By comparing sample to ladder
    - Can infer
        - quantity of base pairs in each sample
        - (how long fragments are)
- 

4. Visualising dna fragments
- Add dye to stain fragments
    - As dna is invisible
- Makes us see fragments
- When to be added:
    - To agarose liquid
        - Before put into mould
    - Or to gel
        - After being run

Applications of PCR and Gel Electrophoresis:

- Dna sequencing
    - Determining base sequence of dna
    - Use pcr to amplify dna

- Sanger reaction to make
  - fluorescent fragments
    - For sequencing
- Run fragments on electrophoresis gel
  - To determine sequence
- Dna profiling
  - analysing dna variations
    - For identification
  - Pcr to amplify short tandem repeats (STRs)
    - Str lengths
      - vary between individuals
        - So they can be used to make dna profiles
  - Run fragments on electrophoresis gel
    - To determine sizes of str fragments

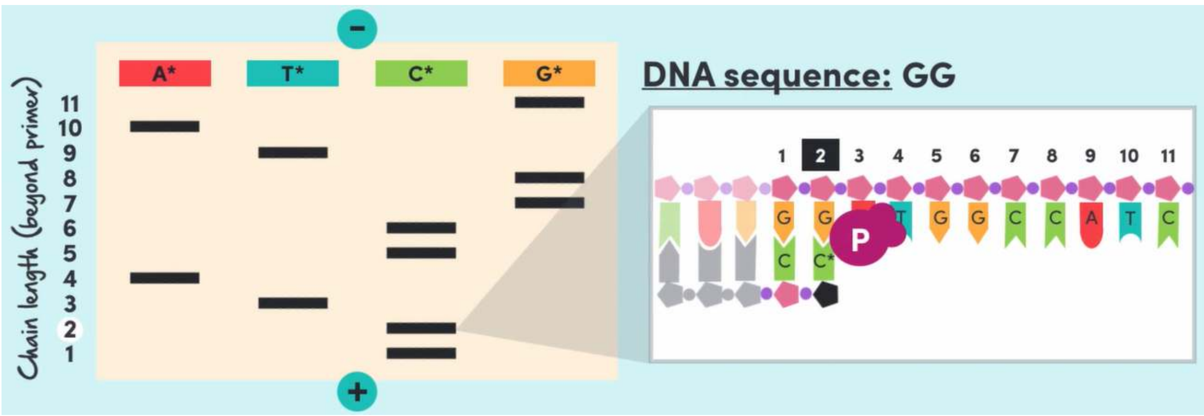
DNA Sequencing:

- Exact nucleotide sequence (order of bases)
  - On chromosome
    - Is determined
- Focus on
  - Full genetic code of
    - Gene or
    - Genome
- Used in
  - Genetic research
  - Mutation detection
  - Medicine
- Methods
  - Sanger
  - Maxam gilbert
- Very detailed
- Pcr and gel electrophoresis
  - May be used to do this

Gel Electrophoresis:

Process:

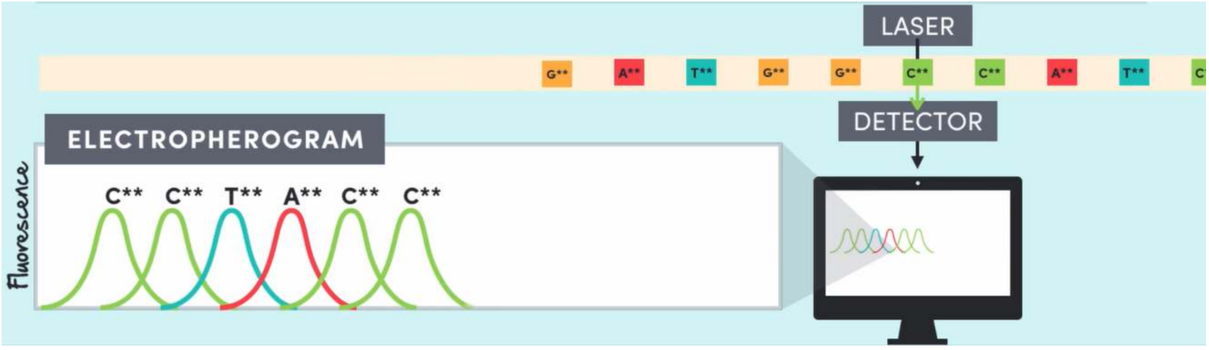
1. Collect dna sample
- Gotten from anything
    - that has cells
- E.g.
    - Hair
    - Saliva
    - blood
2. Extract dna from sample
- Chemicals added that
    - Break cells open
- Dna separated from other parts of cell
3. Amplify dna
- If needed
    - If sample was too small
- Using pcr
    - To make lots of copies of dna
4. Start of sanger reaction
- Separately identifies position of each nucleotide
- Uses 4 special pcrs
    - Normal pcrs with
        - chain terminating nucleotide
- Chain terminating nucleotide
    - Normal 4 nucleotides
    - Without hook to bind to next nucleotide
    - Dna synthesis stopped when this added
- Example guanine chain terminating nucleotide
    - Dna made until this added
    - This added to a cytosine
- Reaction repeated many times
    - For all 4 base pcrs
- In all pcrs
    - Produce fragments of different lengths
5. Determine dna sequence
- Run all 4 pcr reactions

    - On electrophoresis gel
    - To determine lengths
        - Of fragments in each reaction
- Position of base determined by
    - Seeing the shortest length base on gel
    - Recognise that it binded to its complementary base
        - E.g. if guanine on gel
        - It binded to cytosine
    - That complementary base is first in sequence
    - Continue for all bases from
        - Shortest → longest
            - From pos → neg on gel box
- 

Capillary Electrophoresis:

Process:

- First 3 steps same as gel
- 4. Sanger sequencing reaction
    - 4 pcrs
    - With fluorescent chain terminating molecule
        - 4 bases 4 different dyes
- 5. Determine dna sequence
    - Run 4 pcrs on capillary electrophoresis gel
    - Capillary tube with gel
        - Instead of gel box
    - Smaller fragments move → end of tube faster
    - End of tube
        - laser
    - As each fragment passes laser

        - Fluorescent tag
            - Lights up
    - Fluorescence measured by detector
        - Recorded to
            - Make electropherogram
    - 

    - Correct sequence determined by
        - Left → right
        - Nucleotides are
            - Complementary to
                - fluorescent chain terminating nucleotide

Applications of DNA Sequencing:

- Genetic testing
    - For risk of genetic disease
    - Testing presence of abnormal gene
        - Related to disease
- Molecular biological research:
    - To find genes
        - That cause
            - specific disorders
    - Learn more about disease
        - Improve
            - Diagnosis
            - Treatment methods
    - Identify drug targets
    - Personalised medicine
- Evolution:
    - Used in Evolutionary biology
        - Determine inheritance patterns
    - Used to study

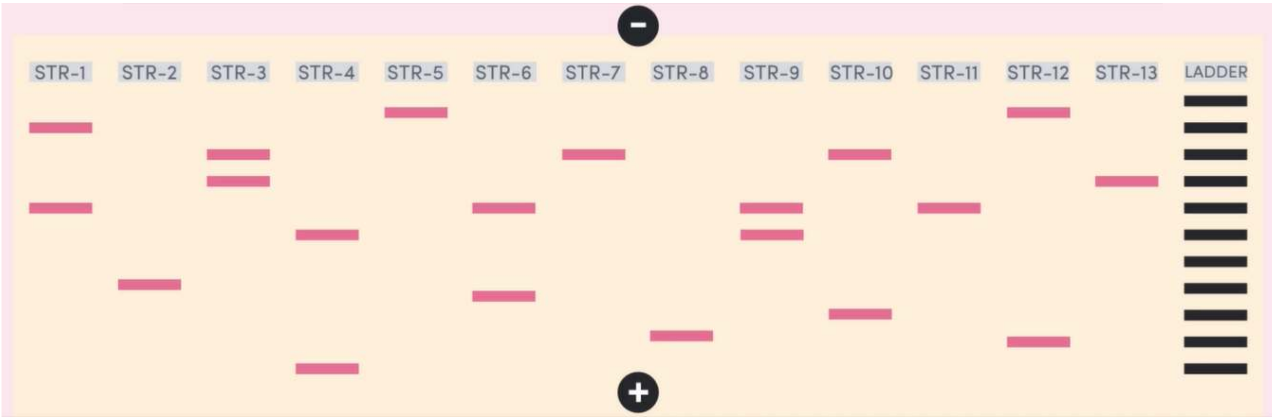
- Different organisms
        - + how they evolved
- Identification:
    - Identify
    - + compare people
    - E.g.:
        - Determine persons biological parents
            - Based on combinations
                - Of their alleles
        - Identity of person
            - At crime scene

DNA Profiling:

- Process of
    - Analysing dna variations
        - For identification of people
- Genetic markers
    - Regions of dna
        - That vary between individuals
- Since known location of gene on chromosome
    - Used to construct
        - dna profiles
- Dna variations:
    - Short Tandem Repeats (STRs)
        - String of repeating
            - Nucleotides
            - (2 - 5 bases long)
        - Many different strs in genome
            - In different places
        - Are non coding dna
            - Don't do protein making
                - ∴ no impact on health
- Strs repeating bases same
    - But number of times repeated
        - Different
            - for different people
    - Results in

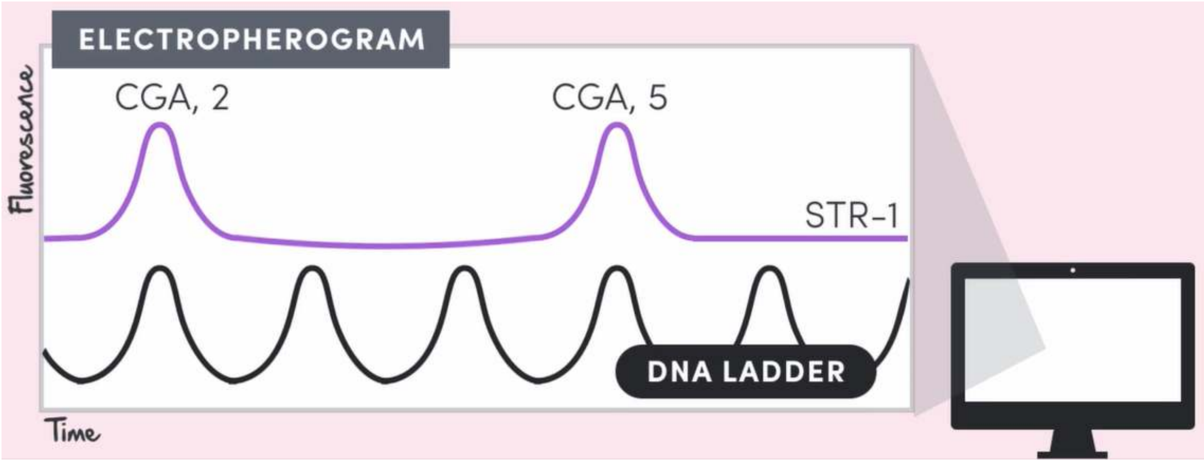
        - Different str lengths
            - Diff no. of nucleotides
- Dna profiles
    - Based on diff in str lengths

Gel Electrophoresis:

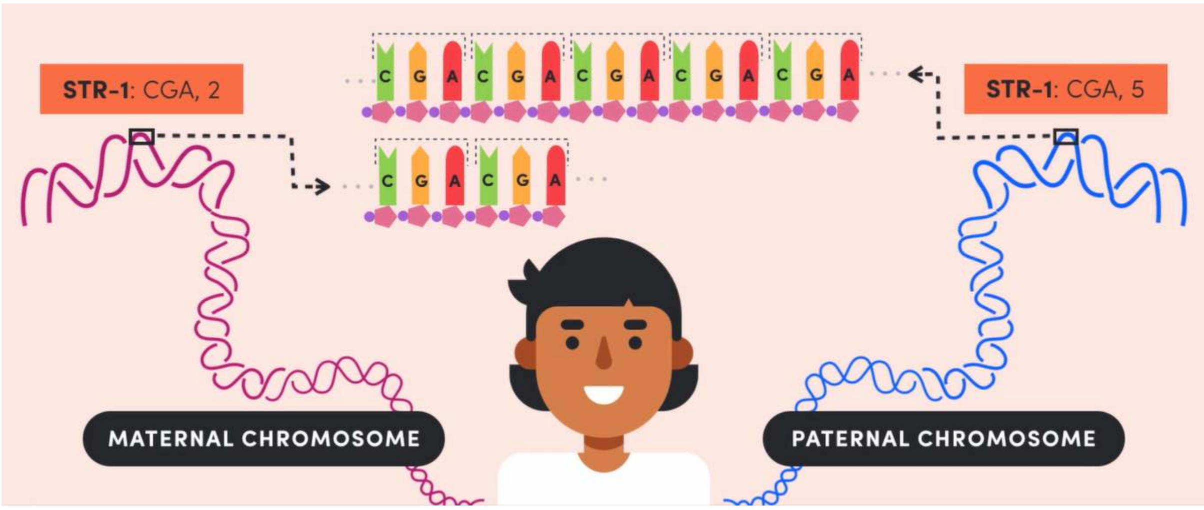
- Same thing as normal dna sequencing
- But only done for strs
    - Instead of individual bases
- To make full dna profile
    - Need 13 strs
- 

Capillary Electrophoresis:

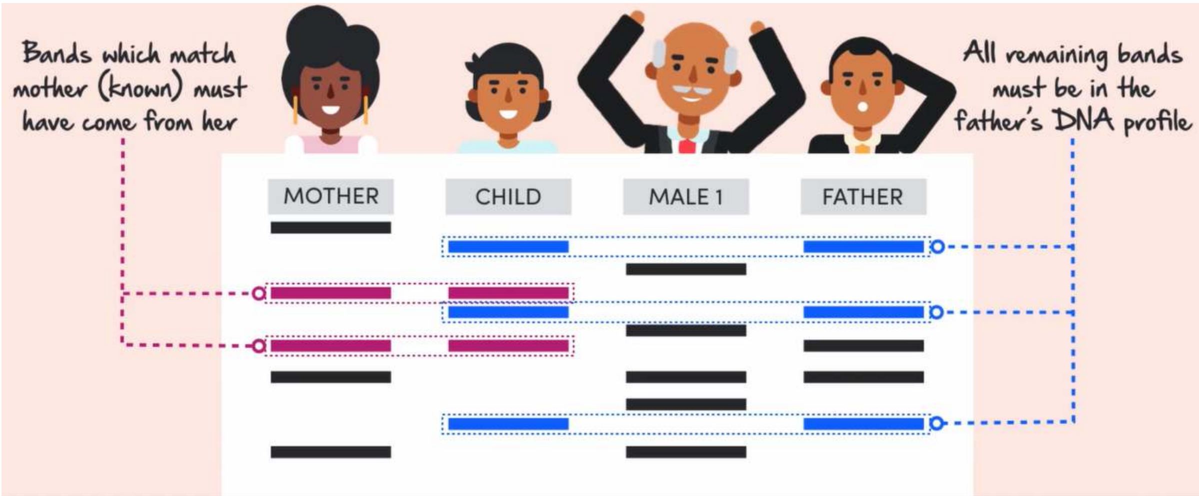
- Same as normal capillary electrophoresis
- But only done for strs
    - Instead of individual bases
- 

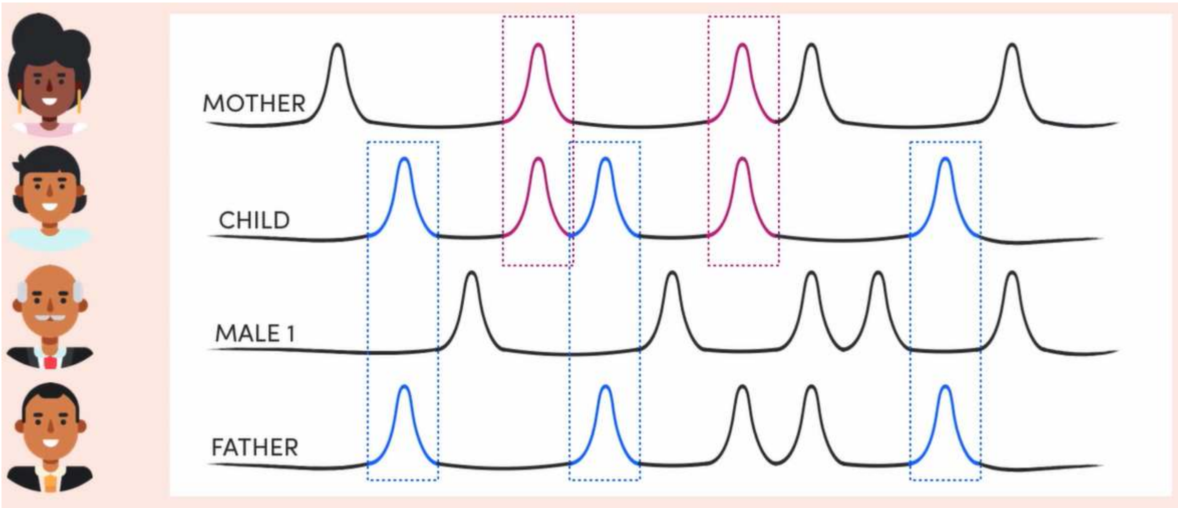
- 

Applications of DNA Profiling:

- Determining parentage:
    - Every person has 2 alleles
    - (copies of every str)
        - In their genome
    - As they have
        - 1 set of chromosomes
            - From each parent
        - 1 from mom
        - 1 from dad
    - 

    - ∴ all str alleles in persons dna prof
        - Should be in
            - Mother +
            - Or fathers
                - Dna profile
    - Child dna profile
        - Bands or peaks in it should match both

- Bands or peaks in mom
- + dads profile
- 

- 

- Identification
    - Determine identity of unknown person
    - Forensic investigations
        - Determine identity of
            - Crime suspect
            - Crime victim
                - If unrecognisable
    - Natural disasters
        - Determine identity of
            - People killed in
                - Natural disasters
                - E.g. earthquakes, tsunamis
    - sample of dna taken from scene
        - Dna profile generated
    - Victims or suspects identified by

- Seeing if their profile
    - Is complete match
        - To the scene profile